Clustering Analysis

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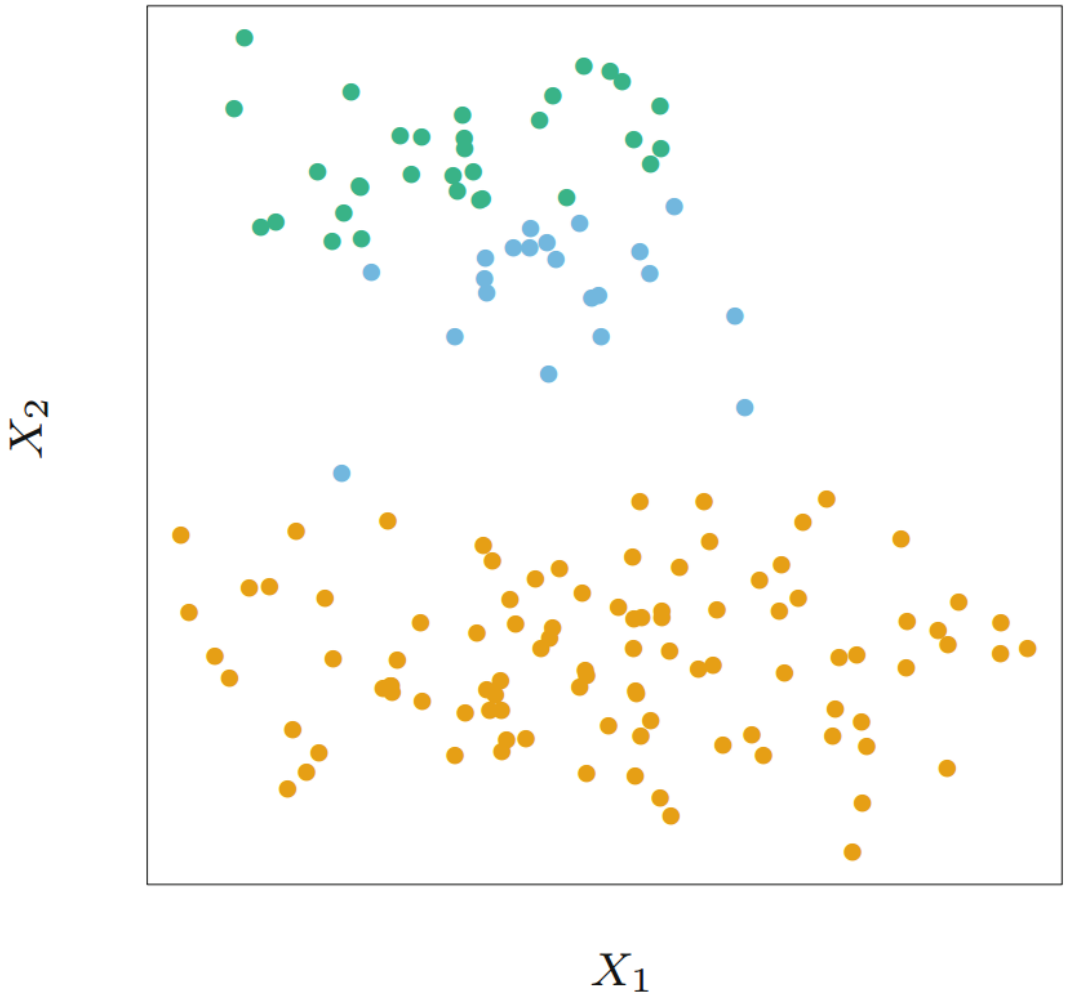
Clustering

- **Clustering** refers to a very broad set of techniques for finding **subgroups**, or **clusters**, in a data set.

- We seek a partition of the data into distinct groups so that the observations within each group are quite **similar** to each other.

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Clustering



Clustering examples

- For example, suppose we have access to a large number of measurements (e.g. household income, occupation, distance from nearest urban area, and so forth) for a large number of people.

- Our goal is to perform **market segmentation** by identifying subgroups of people who might be more receptive to a particular form of advertising, or more likely to purchase a particular product.

- The task of performing market segmentation amounts to clustering the people in the data set.

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Clustering vs. Classification vs. Dimension Reduction

Classification	Clustering	Dimension Reduction
Supervised	Unsupervised	Unsupervised
Classifies samples into predefined classes	Identify groupings from the given samples	Create low dimensional representations that preserve similarity in high-dimensional manifolds
Used for prediction	Applies only to the data at hand	Applies only to the data at hand

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Clustering

- We introduce 4 clustering algorithms:
  - K-means
  - Hierarchical clustering
  - Spectral clustering
  - BIRCH
- Before that, we first need to define **similarity/dissimilarity**.

Dissimilarities Based on Attributes

- Most often we have measurements x_{ij} for $i = 1, \ldots, N$, on variables $j = 1,2, \ldots, p$ (also called attributes).

- In the most common case, we define a dissimilarity $d_j\left(x_{ij}, x_{i'j}\right)$ between values of the jth attribute. For example, the squared distance

$$d_j\left(x_{ij}, x_{i'j}\right) = (x_{ij} - x_{i'j})^2.$$

- Then define

$$D(x_i, x_{i'}) = \sum_{j=1}^{p} w_j \cdot d_j\left(x_{ij}, x_{i'j}\right)$$

as the dissimilarity between objects i and i'.

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Quantitative Variables

- Naturally, the dissimilarity for two quantitative measurements is a monotone-increasing function of the absolute difference.

- Popular choices are the squared distance

$$d(x_i, x_{i'}) = (x_i - x_{i'})^2$$

and the absolute distance

$$d(x_i, x_{i'}) = |x_i - x_{i'}|.$$

The former places more emphasis on larger differences than smaller ones.

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Ordinal Variables

- The values of this type of variables are often represented as contiguous integers, and the realizable values are considered to be an ordered set
  - academic grades (A, B, C, D, F)
  - degree of preference (can't stand, dislike, OK, like, terrific)
  - ranking

- Dissimilarity measures for ordinal variables are generally defined by replacing their M original values with

$$\frac{i-1/2}{M}, i = 1, \ldots, M.$$

They are then treated as quantitative variables on this scale.

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Nominal Variables

- With unordered categorical (also called **nominal**) variables, the degree-of-difference between pairs of values must be delineated explicitly. If the variable assumes M distinct values, these can be arranged in a symmetric $M \times M$ matrix with elements $L_{rr'} = L_{r'r}, L_{rr} = 0, L_{rr'} \geq 0$.

- The most common choice is $L_{rr'} = 1$ for all $r \neq r'$, while unequal losses can be used to emphasize some errors more than others.

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Missing Values

- Often, observations have missing values in one or more of the attributes. In such cases, one could
  - Omit the missing values when computing the dissimilarity for a pair of observations.
  - Impute using mean, median, or more advanced methods.
  - For categorical variables, one could consider the value "missing" as just another categorical value, if that makes sense for the problem.

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Object Dissimilarity

- Next we combine the p-individual attribute dissimilarities $d_j\left(x_{ij}, x_{i'j}\right), j = 1,2,\ldots,p$ into a single overall measure of dissimilarity $D\left(x_i, x_{i'}\right)$ between two objects or observations $\left(x_i, x_{i'}\right)$

$$D\left(x_i, x_{i'}\right) = \sum_{j=1}^{p} w_j \cdot d_j\left(x_{ij}, x_{i'j}\right).$$

- The choice of weights w_j should be **based on subject matter considerations**.

- It is important to realize that setting the weight w_j to the same value for each variable (say, $w_j = 1, \forall j$) does **not** necessarily give all attributes equal influence.

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Object Dissimilarity

- The influence of the *j*th attribute X_j on object dissimilarity depends upon its relative contribution to the average object dissimilarity measure over all pairs of observations in the data set

$$\bar{D} = \frac{1}{N^2}\sum_{i=1}^{N}\sum_{i'=1}^{N} D\left(x_i, x_{i'}\right) = \sum_{j=1}^{p} w_j \cdot \bar{d}_j \,,$$

with

$$\bar{d}_j = \frac{1}{N^2}\sum_{i=1}^{N}\sum_{i'=1}^{N} d_j\left(x_{ij}, x_{i'j}\right)$$

being the average dissimilarity on the *j*th attribute.

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Object Dissimilarity

- Thus, setting $w_j \sim 1/\bar{d}_j$ would give all attributes equal influence in characterizing overall dissimilarity between objects.

- For example, with p quantitative variables and squared distance,

$$\bar{d}_j = \frac{1}{N^2}\sum_{i=1}^{N}\sum_{i'=1}^{N}(x_{ij} - x_{i'j})^2 = 2var_j,$$

where var_j is the sample variance of X_j. Thus, the relative importance of each such variable is proportional to its variance over the data set.

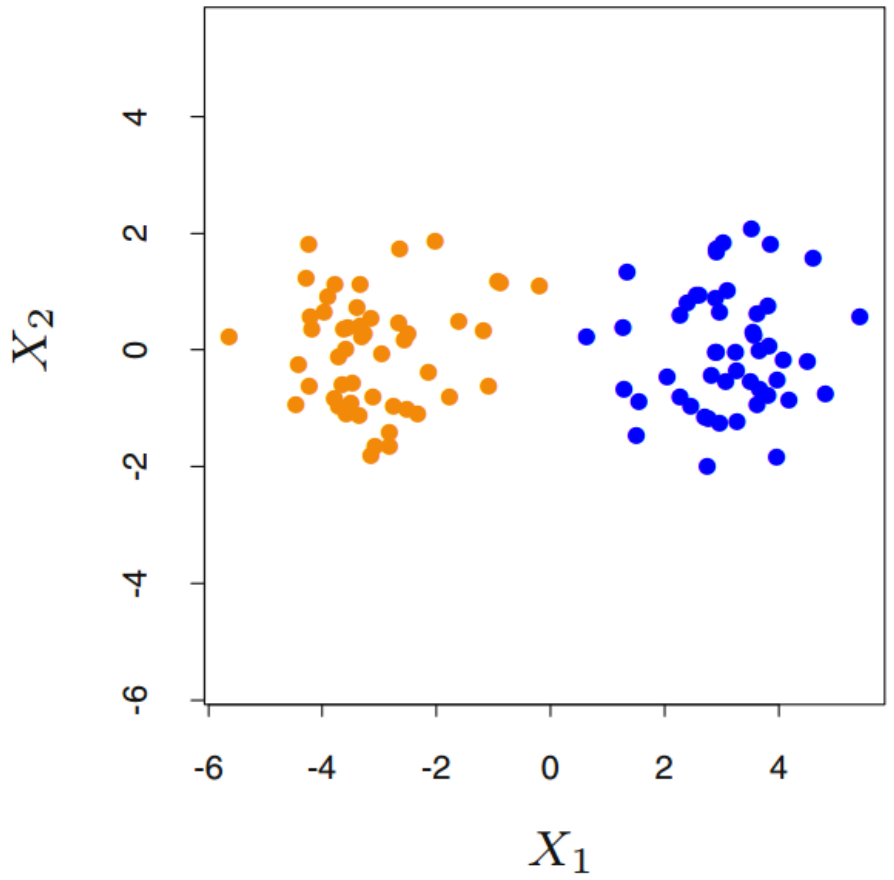
Object Dissimilarity

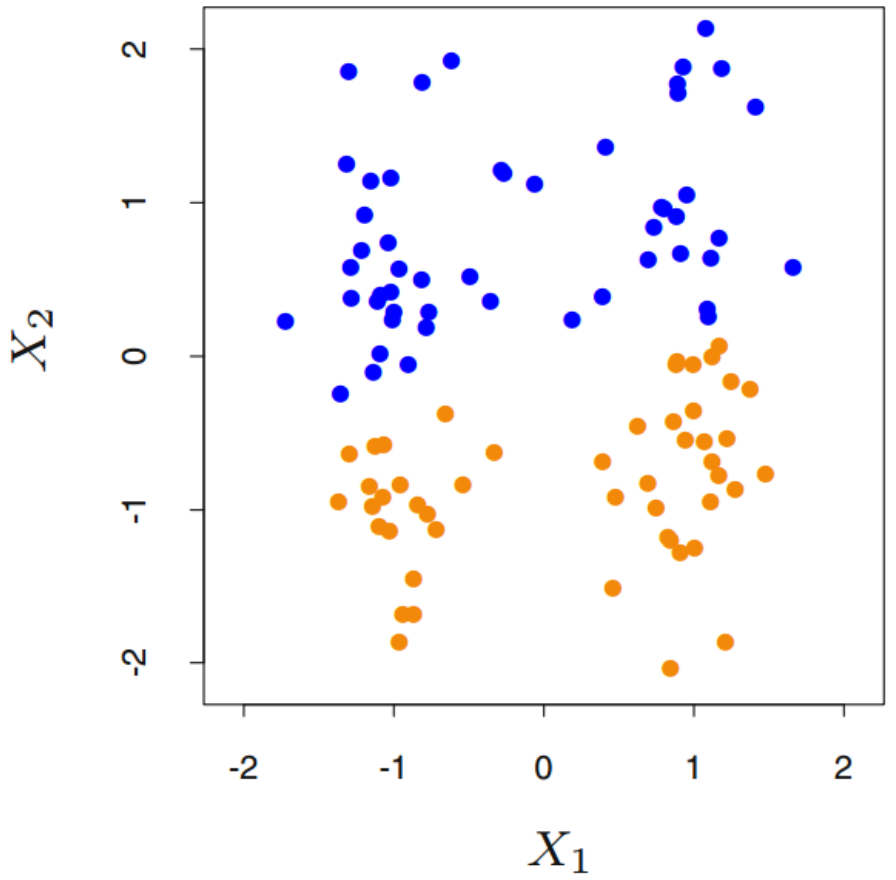
- If the goal is to segment the data into groups of similar objects, **all attributes may not contribute equally** to the (**problem-dependent**) notion of dissimilarity between objects.

- **Some attributes may exhibit more of a grouping tendency than others**. Therefore, giving the variables equal influence may be counterproductive.

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Object Dissimilarity





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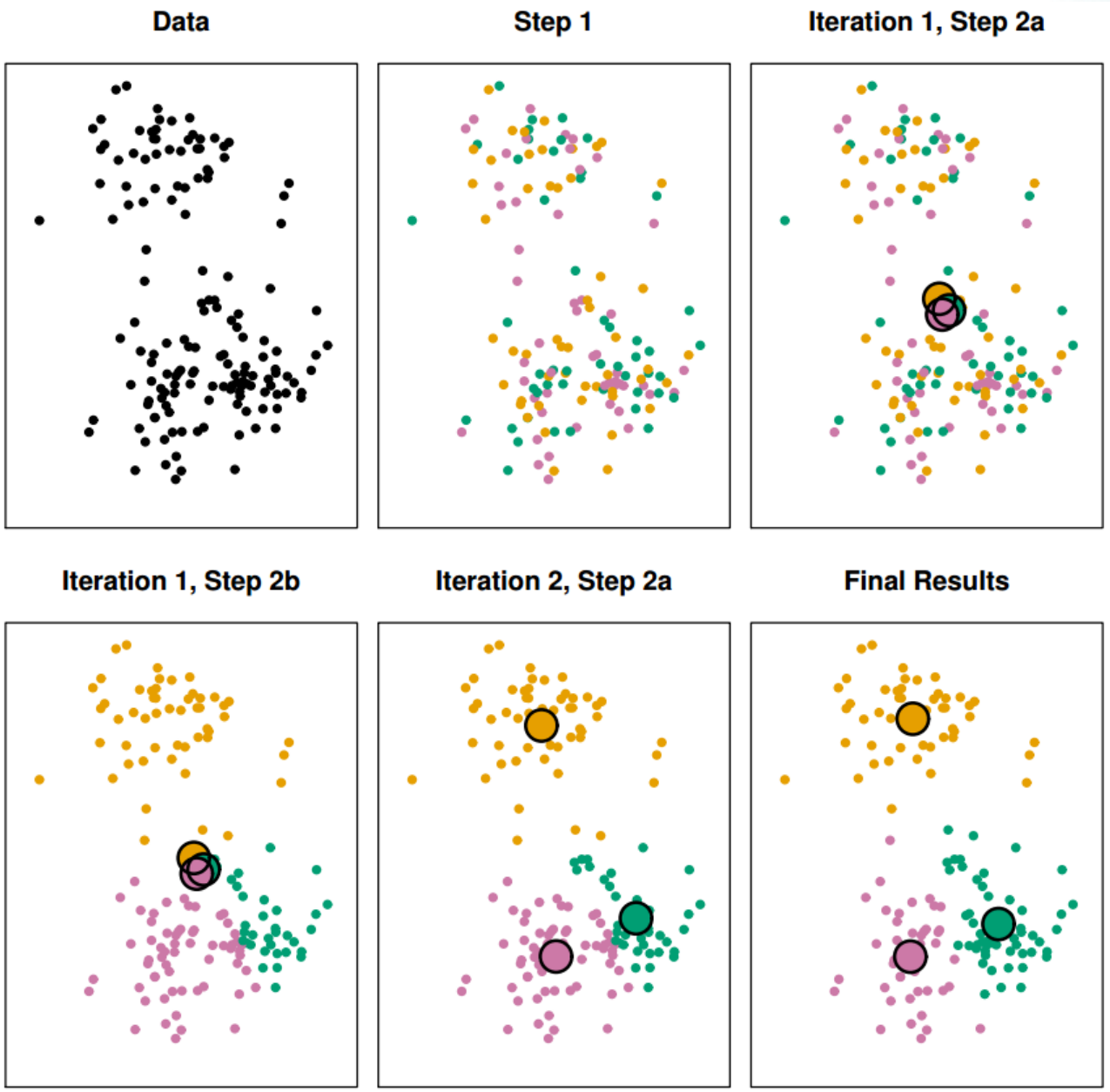
K-means

K-means

- 1. Randomly assign a number, from 1 to K, to each of the observations. These serve as initial cluster assignments for the observations.

- 2. Iterate until the cluster assignments stop changing:
  - 2.1 For each of the K clusters, compute the cluster centroid. The kth cluster centroid is the vector of the p feature means for the observations in the kth cluster.
  - 2.2 Assign each observation to the cluster whose centroid is closest (where closest is defined using Euclidean distance).

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Data
Step 1
Iteration 1, Step 2a
Iteration 1, Step 2b
Iteration 2, Step 2a
Final Results

Trivia

- There are several algorithms for solving the K-means problem.
- The one introduced above is Lloyd's method.
- There are also Hartigan-Wong, MacQueen, etc.
- The default method in R is Hartigan-Wong, which reassign the cluster of one point at a time.

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Example: Human Tumor Microarray Data

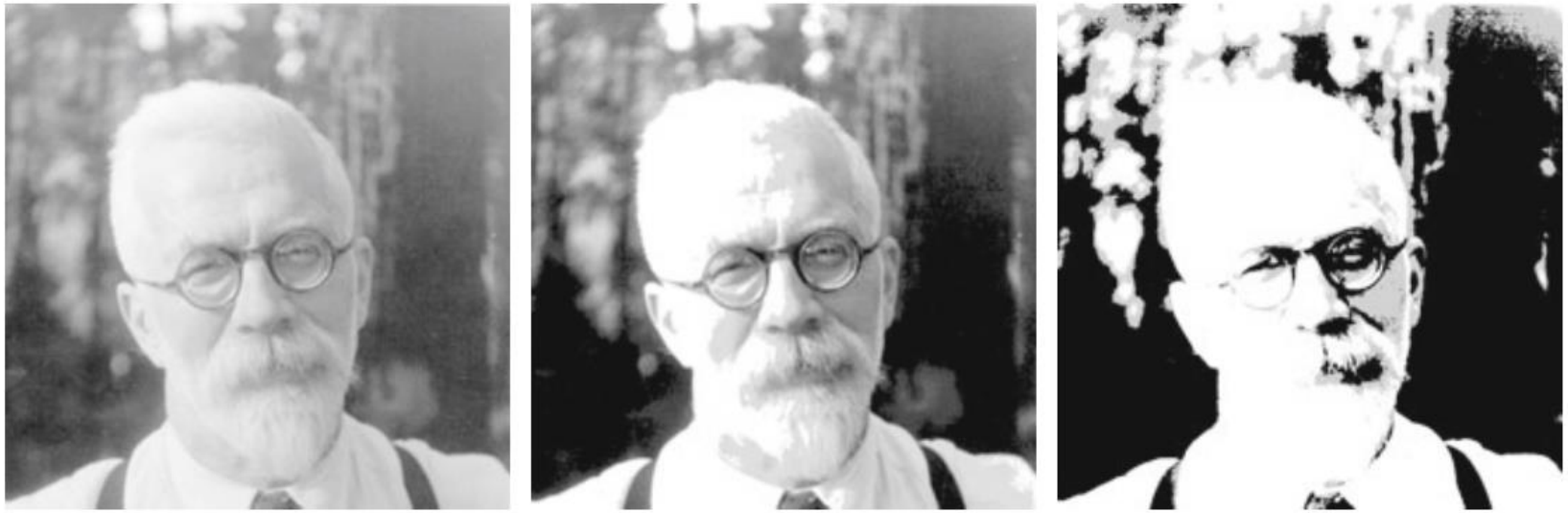
- The data are a 6830x64 matrix of real numbers, each representing an expression measurement for a gene (row) and sample (column).

Cluster	Breast	CNS	Colon	K562	Leukemia	MCF7
1	3	5	0	0	0	0
2	2	0	0	2	6	2
3	2	0	7	0	0	0
Cluster	Melanoma	NSCLC	Ovarian	Prostate	Renal	Unknown
1	1	7	6	2	9	1
2	7	2	0	0	0	0
3	0	0	0	0	0	0

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Application: Vector Quantization

- This is Sir R. A. Fisher

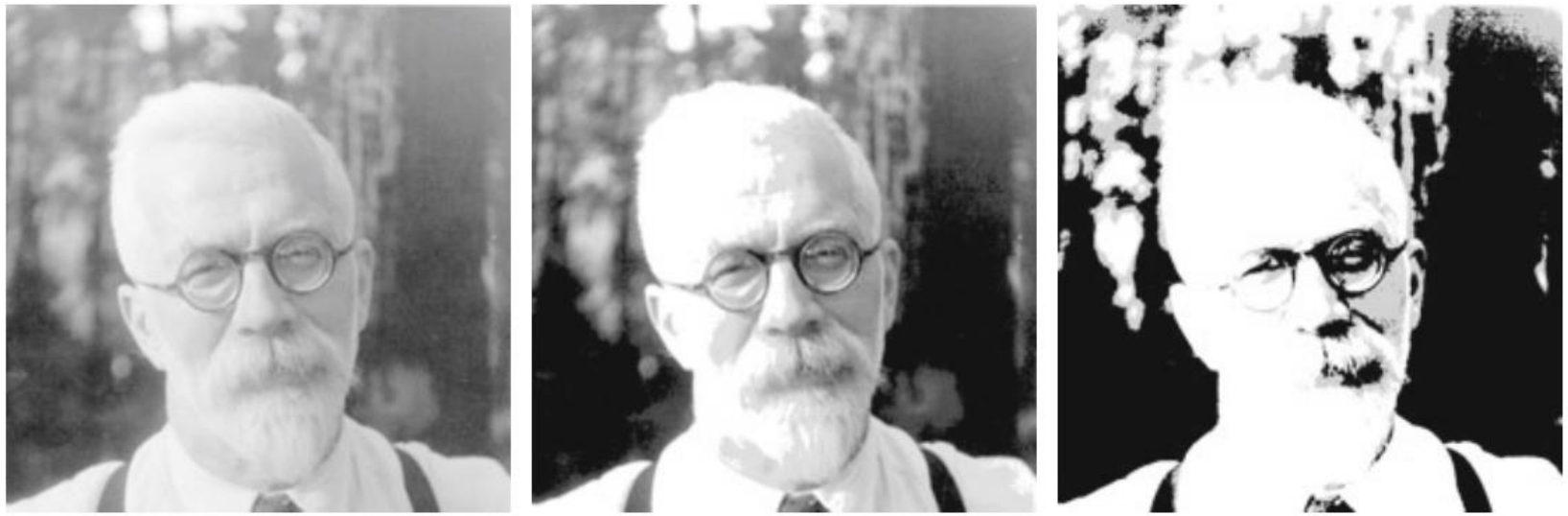


Application: Vector Quantization

- Not the Colonel



Application: Vector Quantization



- Original: 1024x1024 pixels, 0-255 grayscale.
- Every 2x2 pixels form a block, giving 512x512 vectors in $\mathbb{R}^4$.
- Apply K-means as a compression method. Middle: $K = 200$. Right: $K = 4$.

Application: Vector Quantization



Application: Vector Quantization



Application: Vector Quantization



Application: Vector Quantization



Paint by numbers (数字油画)





Paint by numbers (数字油画)





Weaknesses of K-means

- K must be specified
- Sensitivity to initial points
- Local minima

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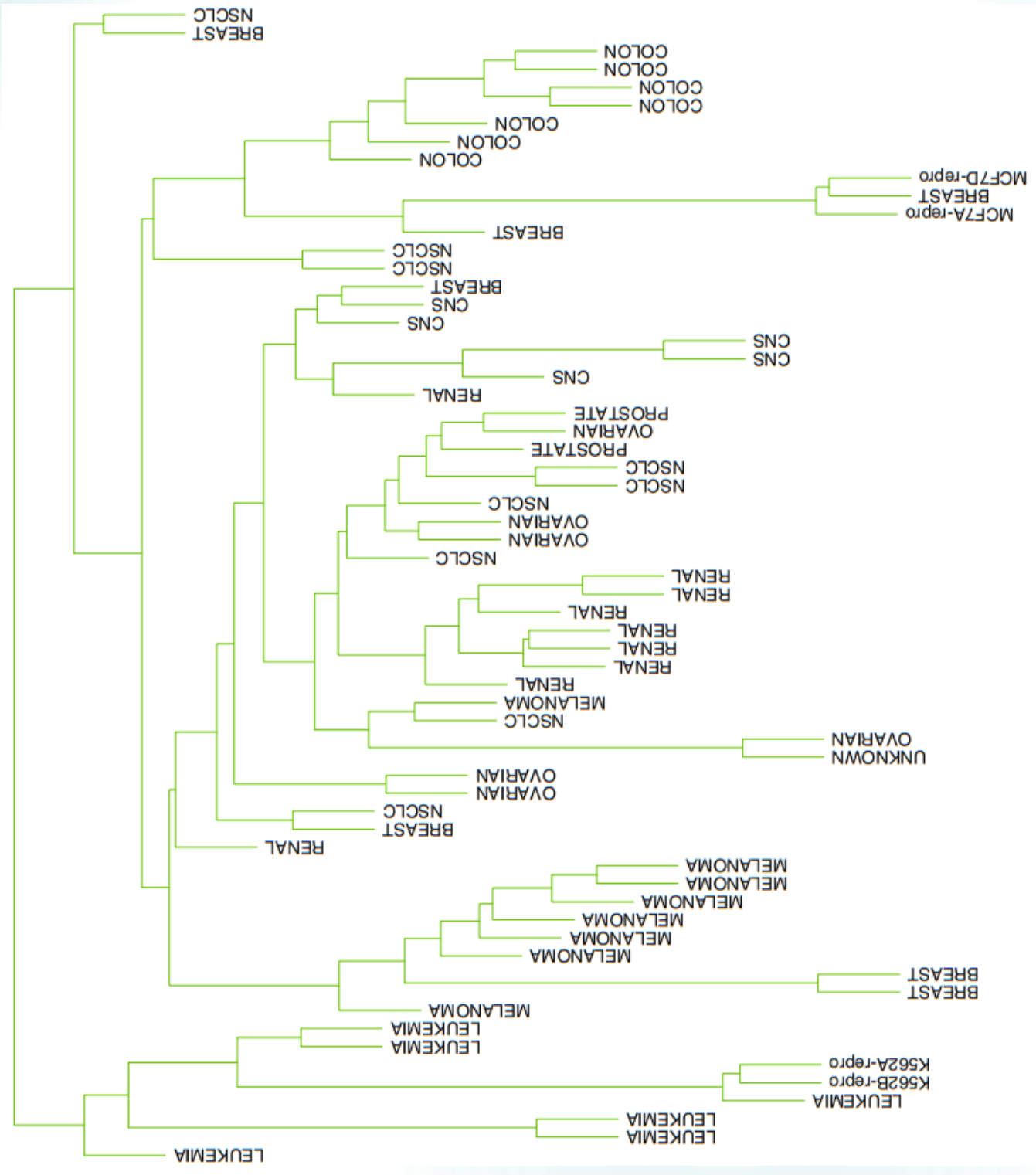
Hierarchical Clustering

Hierarchical Clustering

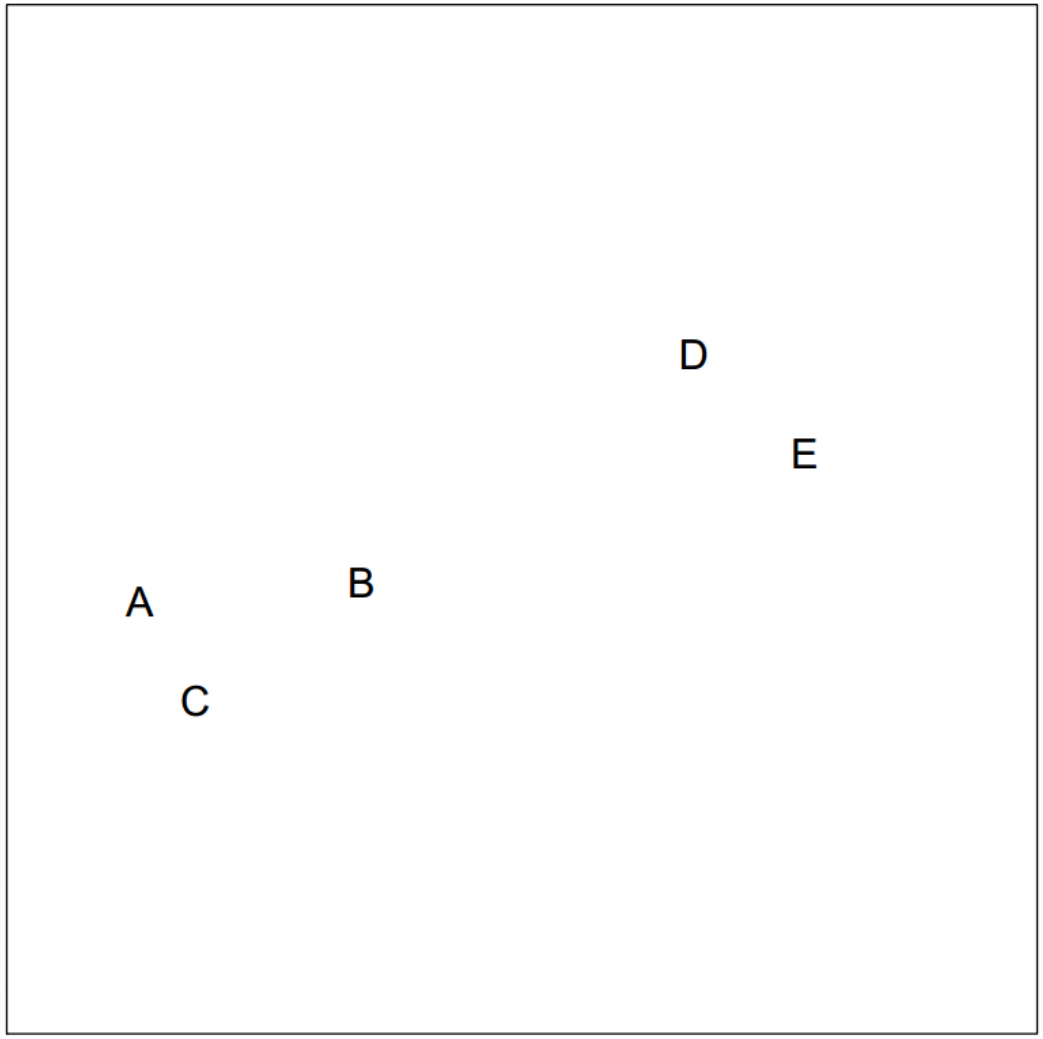
- Hierarchical clustering does not require specifying the number of clusters. Instead, they require the user to specify a measure of dissimilarity between (disjoint) *groups* of observations, based on the pairwise dissimilarities among the observations in the two groups.

- **Bottom-up** or **agglomerative** clustering is the most common type of hierarchical clustering, and refers to the fact that a **dendrogram** is built starting from the leaves and combining clusters up to the trunk.

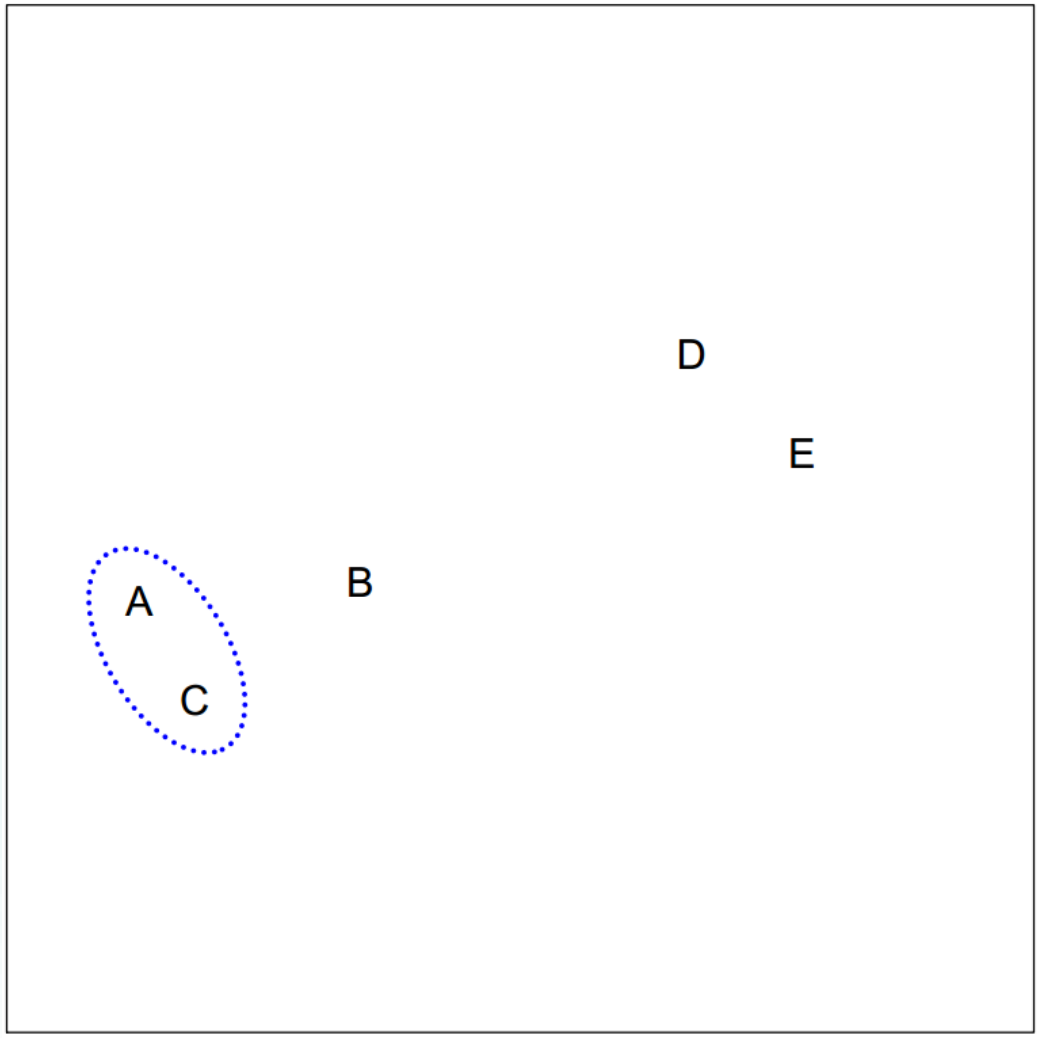
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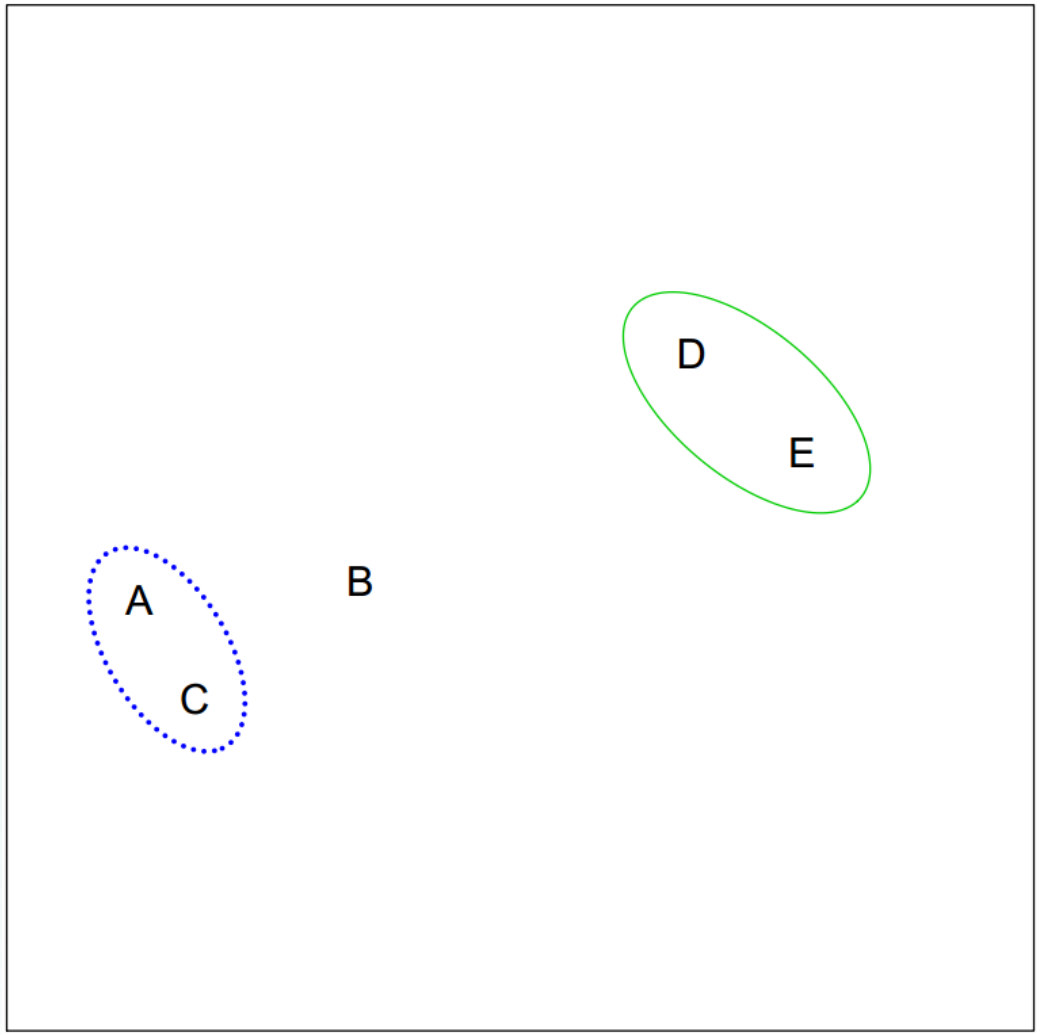
Hierarchical Clustering



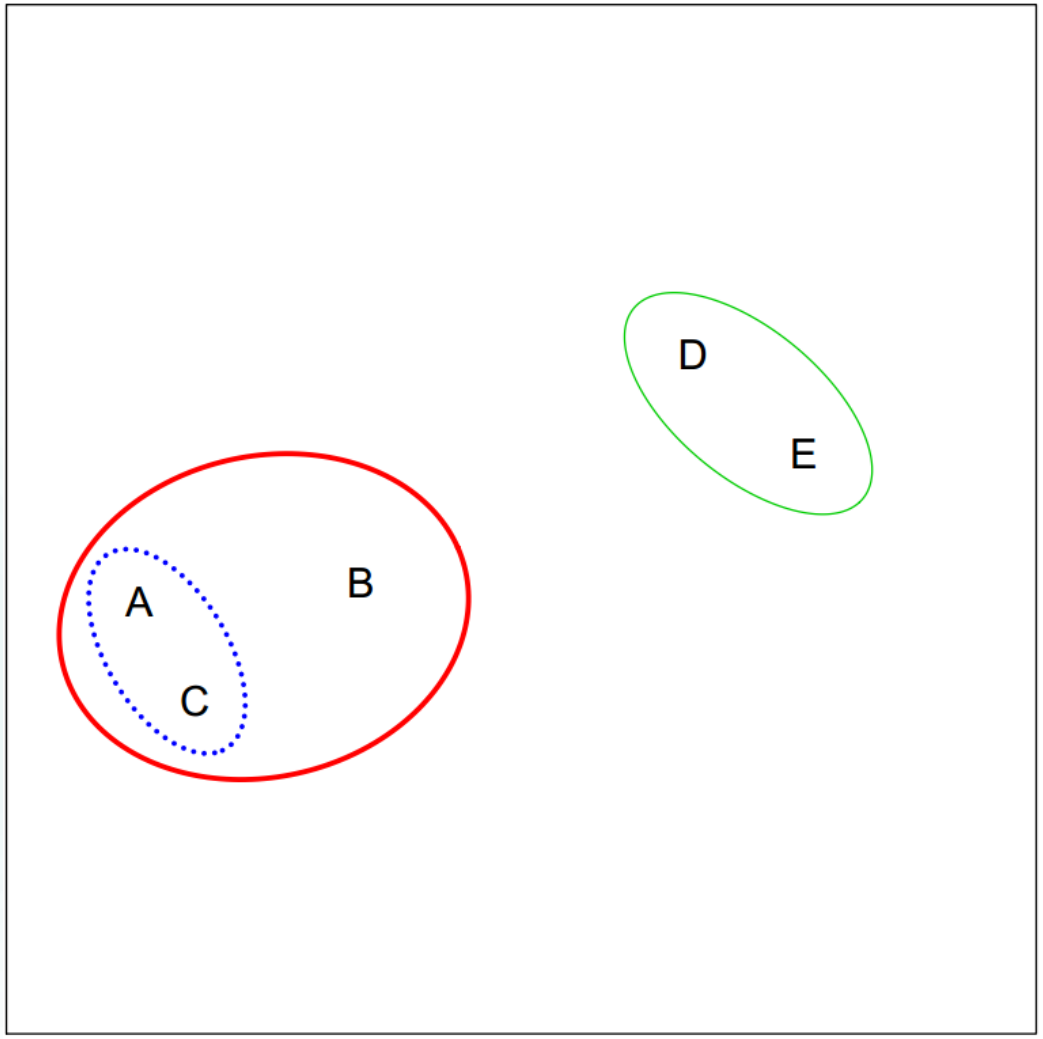
Hierarchical Clustering



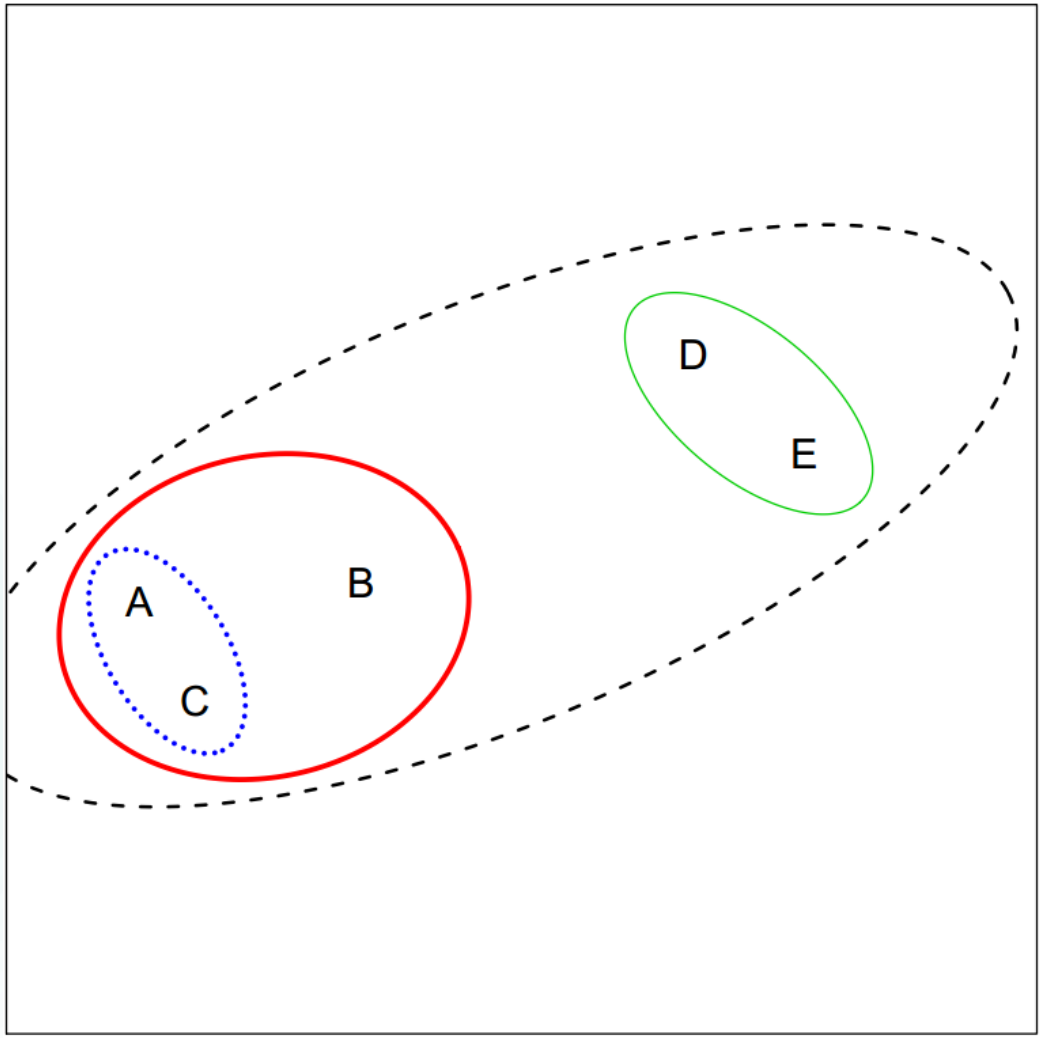
Hierarchical Clustering



Hierarchical Clustering



Hierarchical Clustering

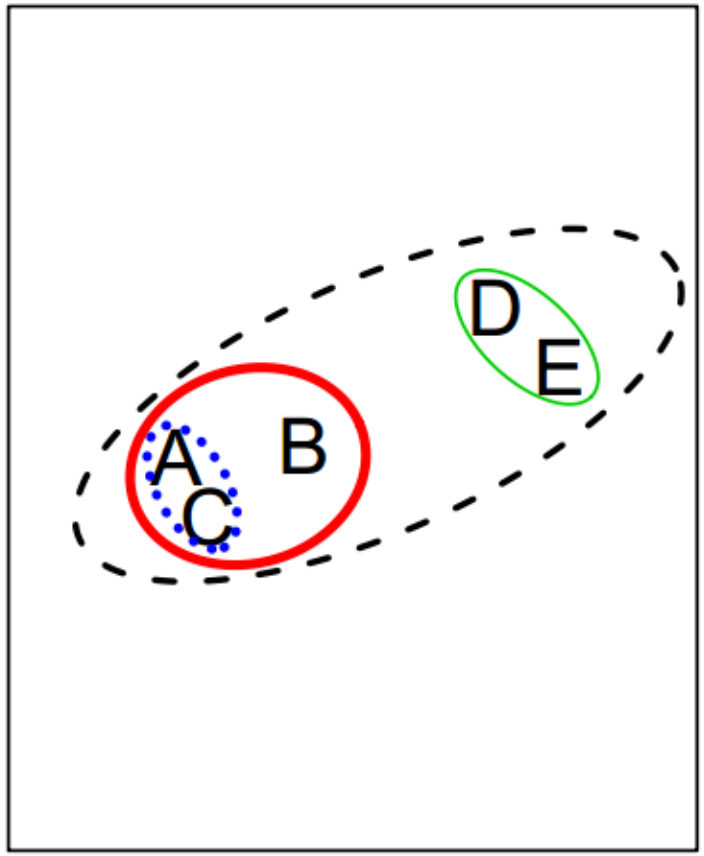


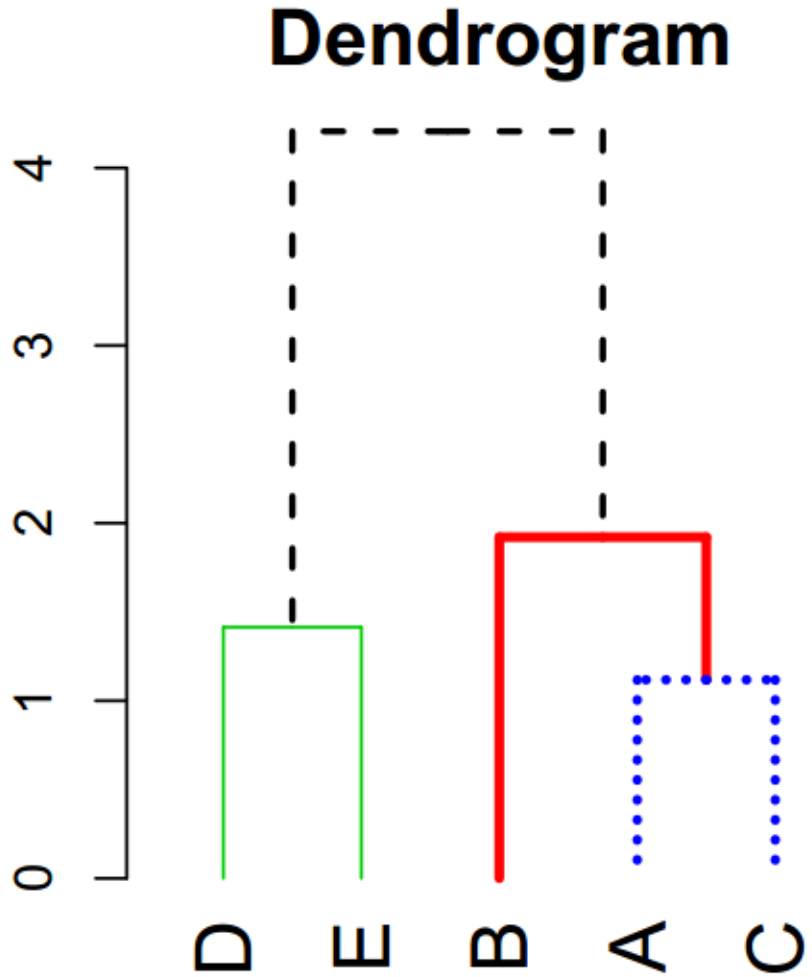
Hierarchical Clustering

- The approach in words:
  - Start with each point in its own cluster.
  - Identify the *closest* two clusters and merge them.
  - Repeat until all points are in a single cluster.

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Hierarchical Clustering





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Intergroup Dissimilarity

- There are three common formulae for the intergroup dissimilarity:
  - Single linkage: $d_{SL}(G,H) = \min_{i\in G, i'\in H} d_{ii'}$.
  - Complete linkage: $d_{CL}(G,H) = \max_{i\in G, i'\in H} d_{ii'}$.
  - Group average: $d_{GA}(G,H) = \frac{1}{N_G N_H}\sum_{i\in G}\sum_{i'\in H} d_{ii'}$.

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Intergroup Dissimilarity

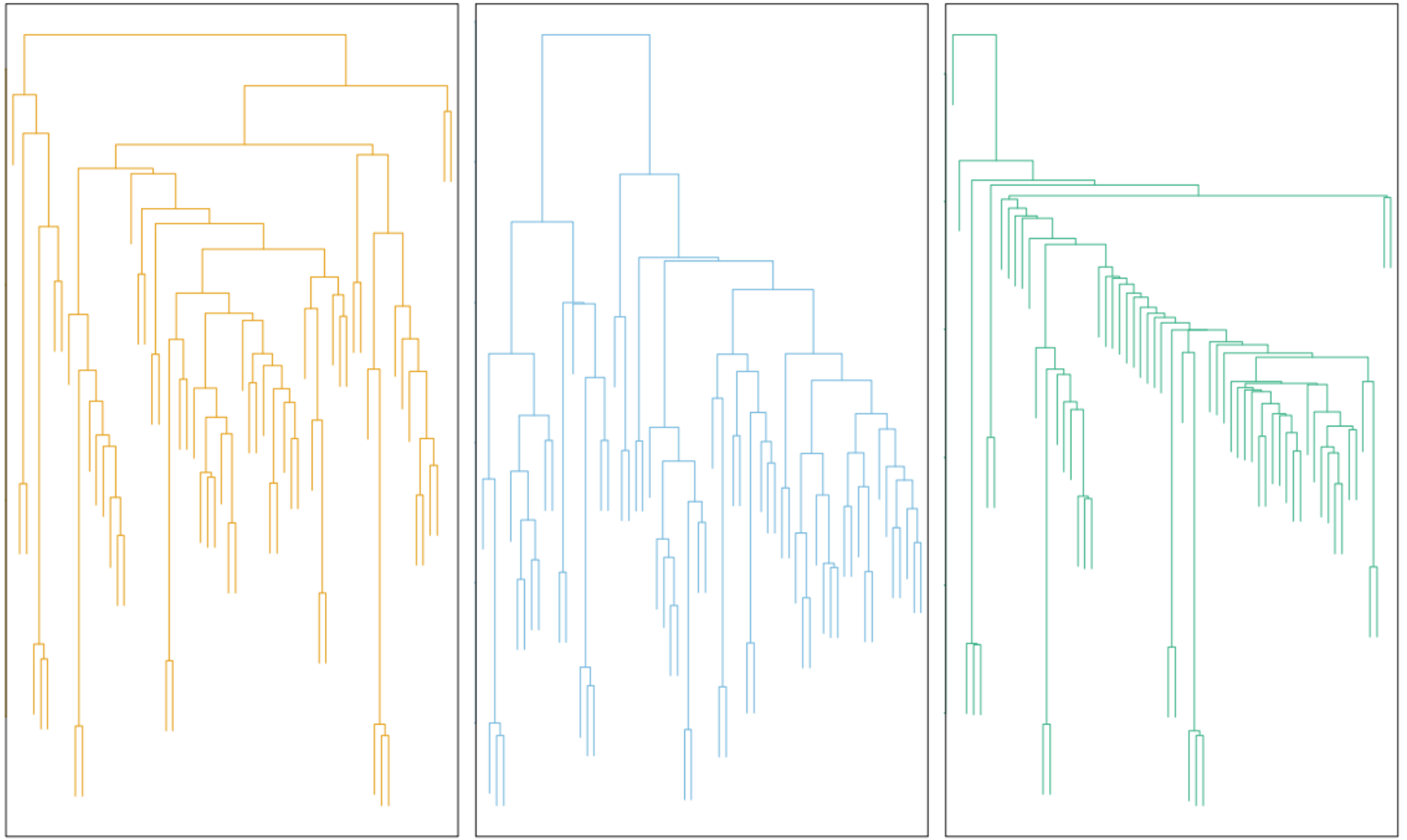
- *Single linkage* only requires that a single dissimilarity $d_{ii'}$ be small for two groups G and H to be considered close together, causing a phenomenon called **chaining**.

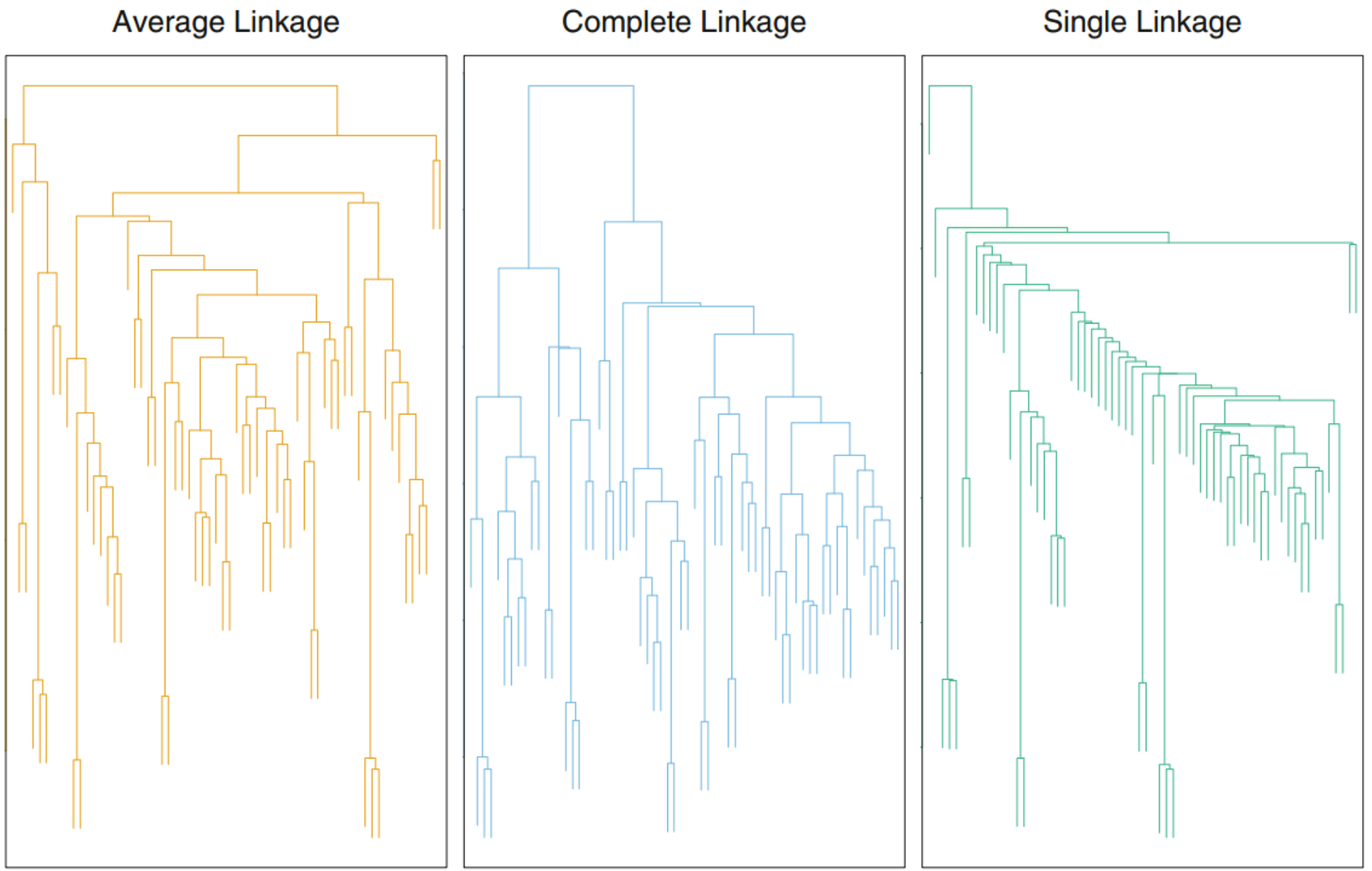
- *Complete linkage* represents the opposite extreme, and tend to produce compact clusters with small diameters. However, it can produce clusters that violate the "closeness" property. That is, observations assigned to a cluster can be much closer to members of other clusters than they are to some members of their own cluster.

- *Group average* represents a compromise between single linkage and complete linkage.

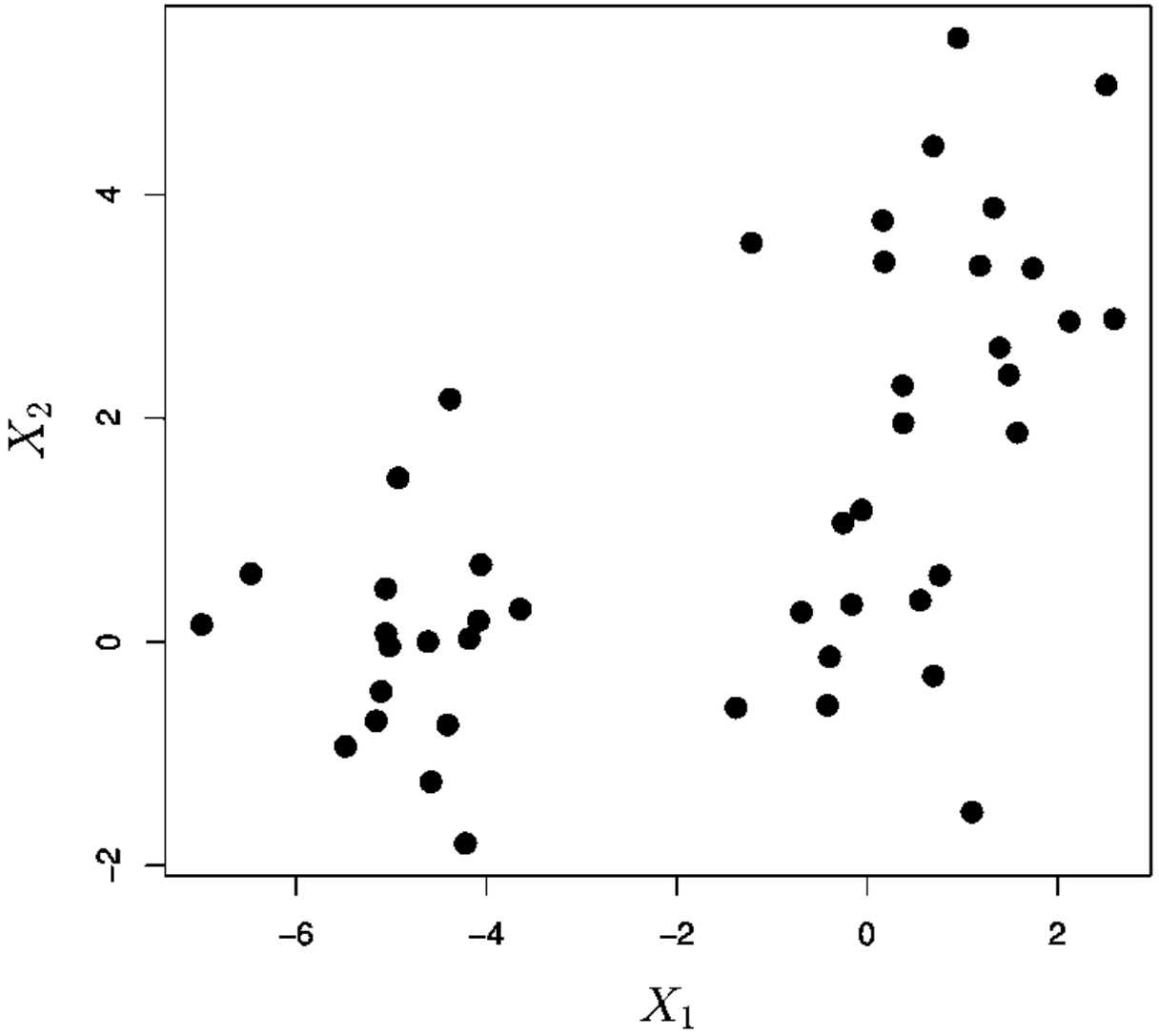
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Which dissimilarity metric does each dendrogram result from?

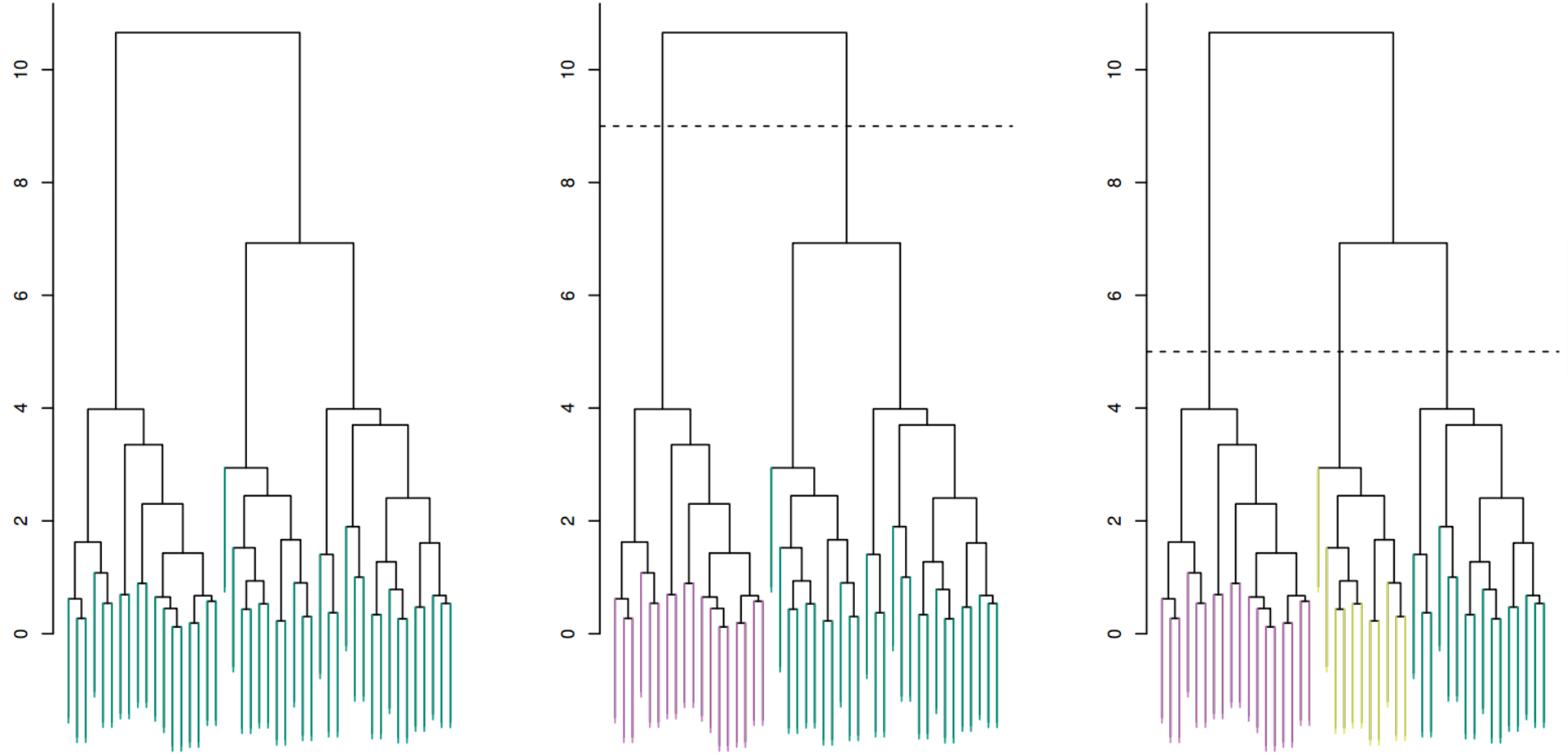



Average Linkage
Complete Linkage
Single Linkage

Determine the number of clusters



Determine the number of clusters



- The y-axis is the intergroup dissimilarity at merging.

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Determine the number of clusters

- Left: Dendrogram obtained from hierarchically clustering the data from previous slide, with complete linkage and Euclidean distance.
- Center: The dendrogram from the left-hand panel, cut at a height of 9 (indicated by the dashed line). This cut results in two distinct clusters, shown in different colors.
- Right: The dendrogram from the left-hand panel, now cut at a height of 5. This cut results in three distinct clusters, shown in different colors.

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Spectral Clustering

Ref: Ulrike von Luxburg. A Tutorial on Spectral Clustering. *Stat Comput* 2007;**17**:395–416.

Spectral Clustering

- Undirected weighted graph $G = (V, E)$.
- Each vertex v_i represents a data point x_i.
- Each edge carries a weight $w_{ii'} = w_{i'i} \geq 0$: $w_{ii'} = 0 \Rightarrow$ vertices v_i and $v_{i'}$ are not connected.
- Clustering ⇒ Graph partition: find a partition of the graph such that edges between clusters have low weights and edges within clusters have large weights.

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ε-neighborhood Graph

- v_i and $v_{i'}$ are connected $\Leftrightarrow \left\|x_i - x_{i'}\right\| \leq \varepsilon$.
- $w_{ii'} = 1$.

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K-nearest Neighbor Graph

- v_i and $v_{i'}$ are connected if x_i is among the k-nearest neighbors of $x_{i'}$ *or/and* if $x_{i'}$ is among the k-nearest neighbors of x_i.
- $w_{ii'} = \exp\left(-\left\|x_i - x_{i'}\right\|^2/\sigma^2\right)$.

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Fully Connected Graph

- Connect all vertices $v_i, v_{i'}$.
- $w_{ii'} = \exp\left(-\left\|x_i - x_{i'}\right\|^2/\sigma^2\right)$.

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Graph Laplacian

- Degree of a vertex v_i is

$$d_i = \sum_{i'=1}^{N} w_{ii'} .$$

- The degree matrix is the diagonal matrix

$$D = diag(d_1, \dots, d_N).$$

- The *unnormalized* graph Laplacian

$$L = D - W.$$

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Properties

- For every $f \in \mathbb{R}^N$,

$$f^T L f = \frac{1}{2}\sum_{i,i'=1}^{N} w_{ii'}\left(f_i - f_{i'}\right)^2 .$$

- L is symmetric and positive semi-definite.
- L has N non-negative, real-valued eigenvalues.
- The smallest eigenvalue of L is 0, and the corresponding eigenvector is the constant One vector.

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Connected Component

- Definition: a subset $A \subset V$ is **connected** if any two vertices in A can be joined by a path such that all intermediate points also lie in A; a subset A is called a **connected component** if it is connected and if there are no connections between vertices in A and A^c; the nonempty sets $A_1, \dots, A_K$ form a **partition** of the graph if $A_k \cap A_{k'} = \emptyset$ and $A_1 \cup \cdots \cup A_K = V$.

- Given a subset $A \subset V$, define the indicator vector

$$\mathbf{1}_A = (f_1, \dots, f_N)^T,$$

where $f_i = 1$ if $v_i \in A$ and $f_i = 0$ if $v_i \notin A$.

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Proposition

- The multiplicity of the eigenvalue 0 of L equals the number of connected components $A_1, \ldots, A_K$ in the graph. The eigenspace of eigenvalue 0 is spanned by the indicator vectors $\mathbf{1}_{A_1}, \ldots, \mathbf{1}_{A_K}$ of those components.

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Proposition

- *Proof*: We start with the case $K = 1$, that is the graph is connected. Assume that f is an eigenvector with eigenvalue 0. Then we know that

$$0 = f^T L f = \frac{1}{2}\sum_{i,i'=1}^{N} w_{ii'}\left(f_i - f_{i'}\right)^2.$$

Proposition

- Now consider the case of K connected components. Without loss of generality we assume that the vertices are ordered according to the connected components they belong to. In this case, the adjacency matrix W has a block diagonal form, and the same is true for the matrix L:

$$L = \begin{pmatrix} L_1 & & \\ & \ddots & \\ & & L_K \end{pmatrix}$$

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Ideal Case

- There are K connected components (clusters) and K is known.
  - Compute L.
  - Find eigenvectors $u_1, \dots, u_K$ corresponding to the eigenvalue 0.
  - Define $z_i = (u_{i1}, \dots, u_{iK})^T, i = 1, \dots, N$.
  - Find distinct values of z_i.

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Non-ideal Case

- In practice, the connected components may not coincide with clusters. There will be edges connecting points in different clusters, but the weights between clusters should be less than the weights within clusters.

$$L = \begin{pmatrix} large & 0 & \ddots \\ 0 & \ddots & 0 \\ \ddots & 0 & large \end{pmatrix} + \begin{pmatrix} \text{small} \end{pmatrix}$$

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Perturbation Theory

- If we perturb a matrix by another matrix with small entries, then the eigenvalues and eigenvectors of the matrix are perturbed by a corresponding small amount.

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Spectral Clustering Algorithm

- Compute L.
- Determine the K smallest eigenvalues of L, $0 = \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_K$, and corresponding eigenvectors $u_1, \ldots, u_K$.
- Define $z_i = (u_{i1}, \ldots, u_{iK})^T, i = 1, \ldots, N$.
- Cluster $\{z_i\}$ using K-means clustering and assign $\{x_i\}$ to corresponding clusters.

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Normalized Graph Laplacian

- $L = I - D^{-1/2} W D^{-1/2}$.
- $L = I - D^{-1} W$.

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Correspondence with Partition Criteria

- Spectral clustering provides approximate solutions to graph partitions

- $L = D - W \Longrightarrow$ *Ratio Cut*

$$\min_{A_1,\ldots,A_k} \frac{1}{2}\sum_{i=1}^{k} \frac{W(A_i, A_i^c)}{|A_i|}$$

where $|A|$ is the number of vertices in A.

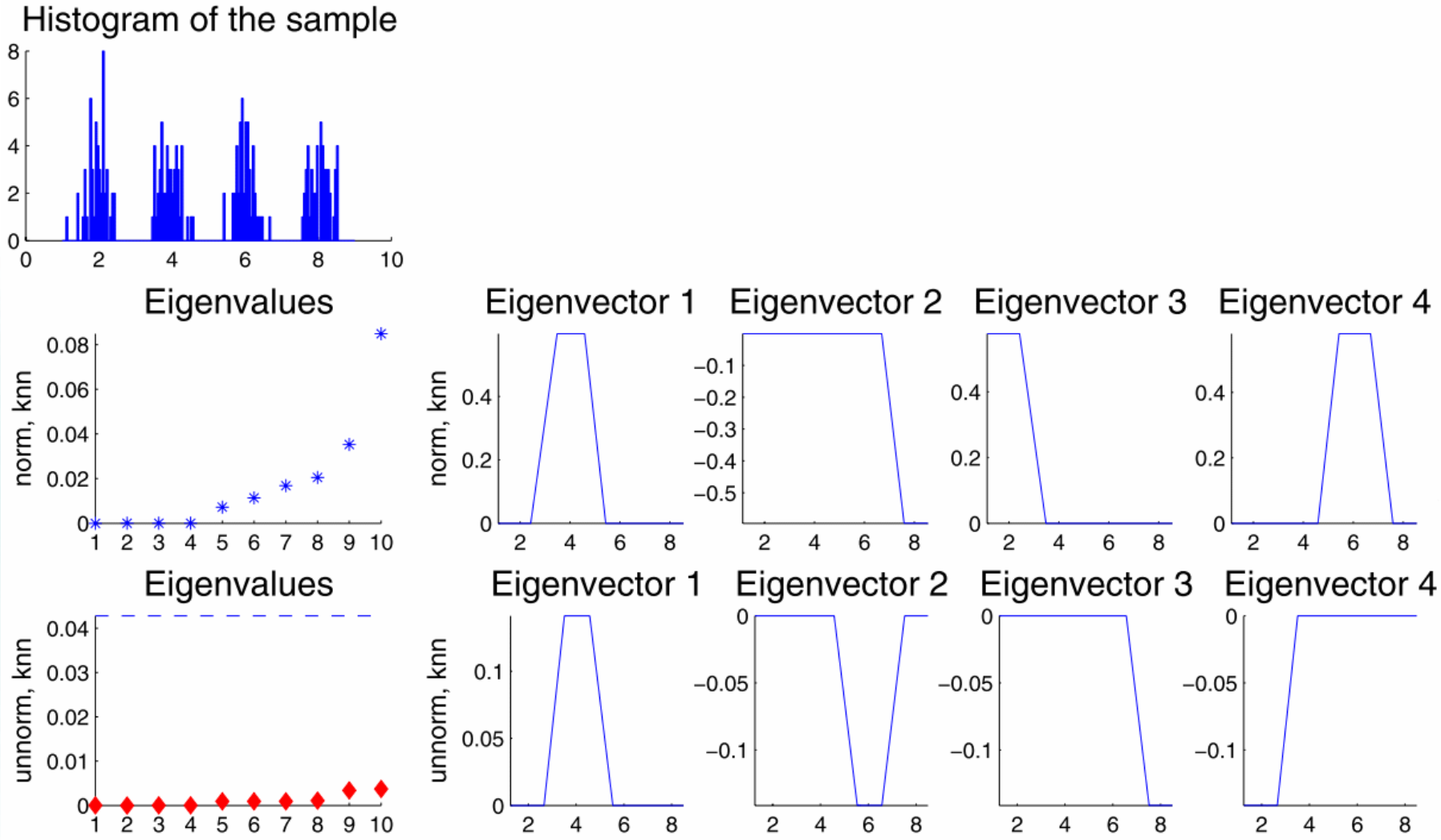
- $L = I - D^{-1/2}WD^{-1/2} \Longrightarrow$ *Normalized Cut*

$$\min_{A_1,\ldots,A_k} \frac{1}{2}\sum_{i=1}^{k} \frac{W(A_i, A_i^c)}{Vol(A_i)}$$

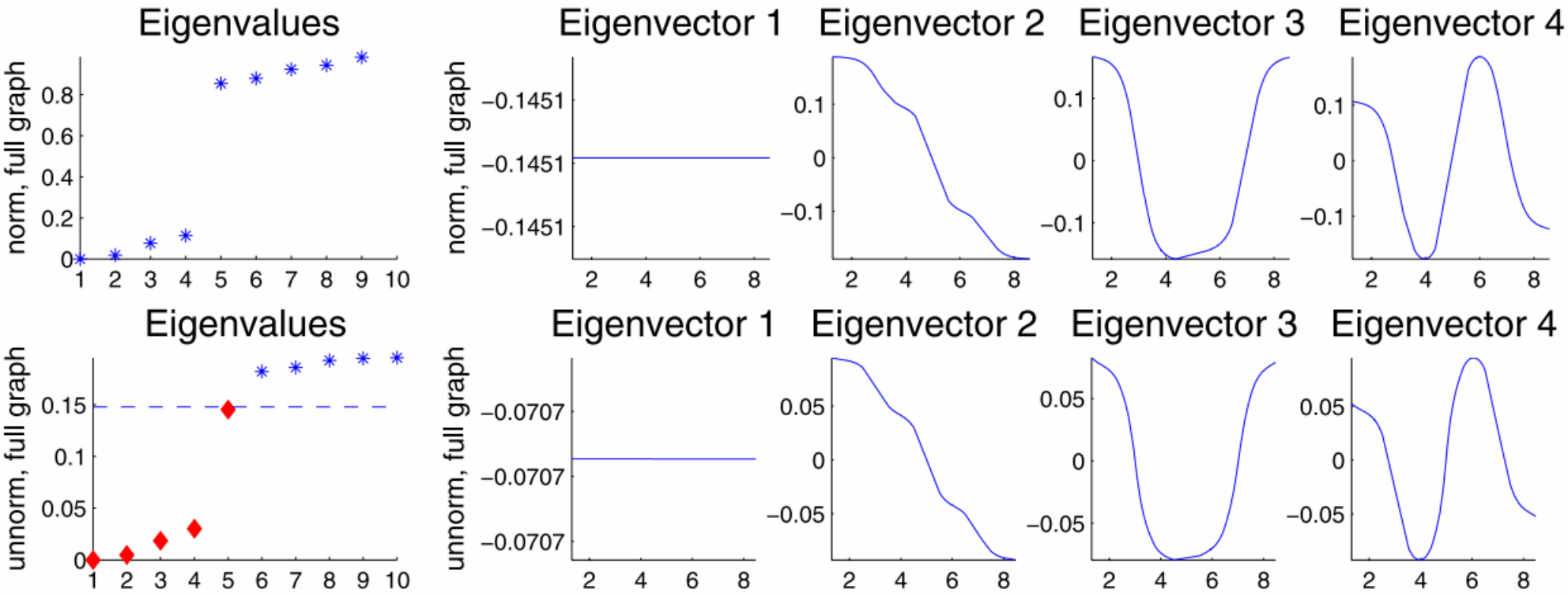
where $Vol(A) = \sum_{i \in A} d_i$.

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Example (k-nn graph)



Example (fully-connected graph)



Ncut and Image Segmentation

- The Normalized Cut (Ncut) algorithm uses the normalized graph Laplacian $L = I - D^{-1/2}WD^{-1/2}$ to bipartition an image.
  - Shi, J., & Malik, J. (2000). Normalized cuts and image segmentation. *IEEE Transactions on pattern analysis and machine intelligence*, *22*(8), 888-905.

- The bipartition is done recursively (recall divisive hierarchical clustering).

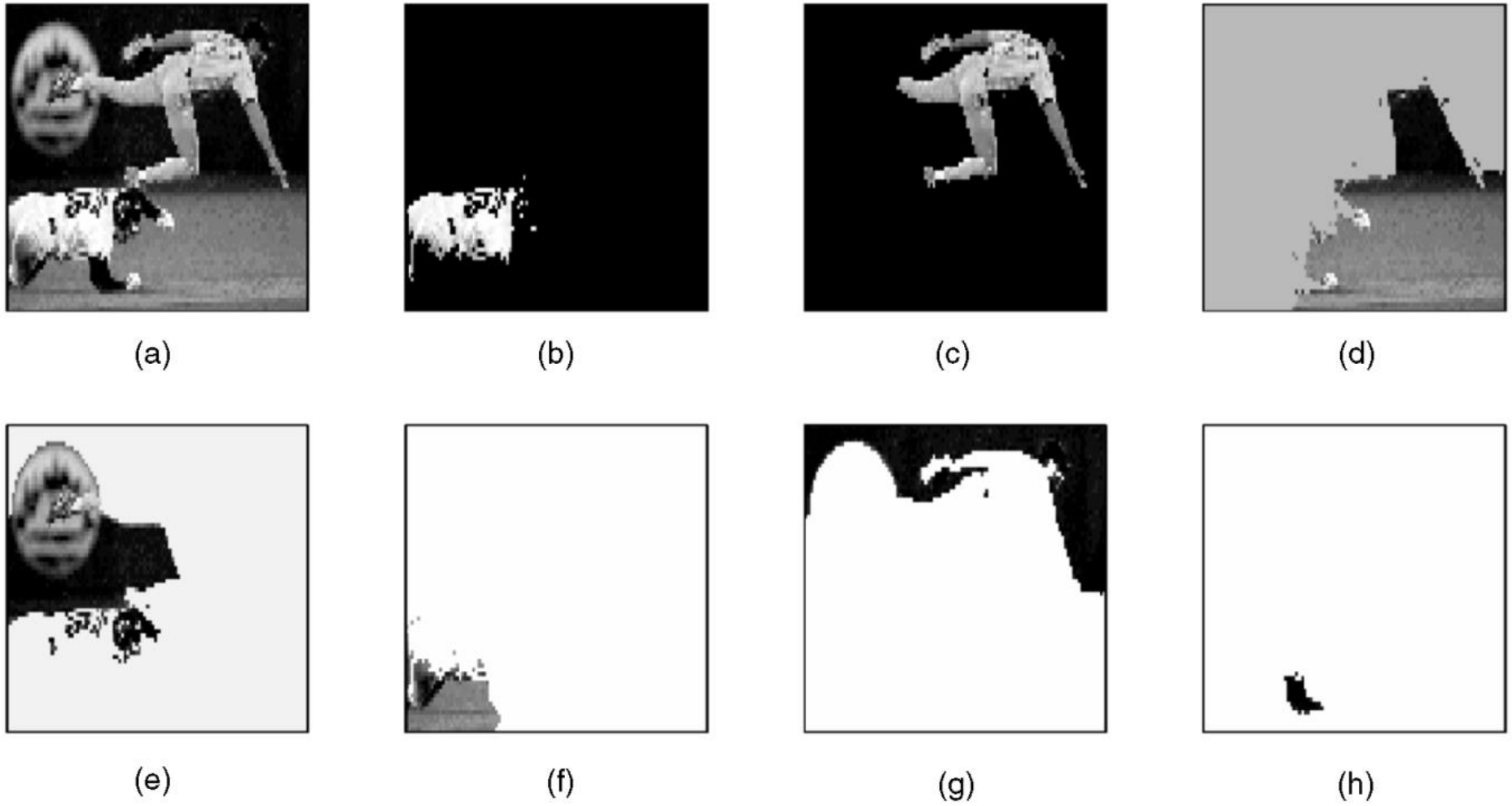
- $w_{ii'} = \exp\left(-\|F(i) - F(i')\|^2/\sigma_I^2\right)$

$$\begin{cases} \exp\left(-\|X(i) - X(i')\|^2/\sigma_X^2\right) & \text{if } \|X(i) - X(i')\| \le \varepsilon \\ 0 & \text{otherwise} \end{cases}$$

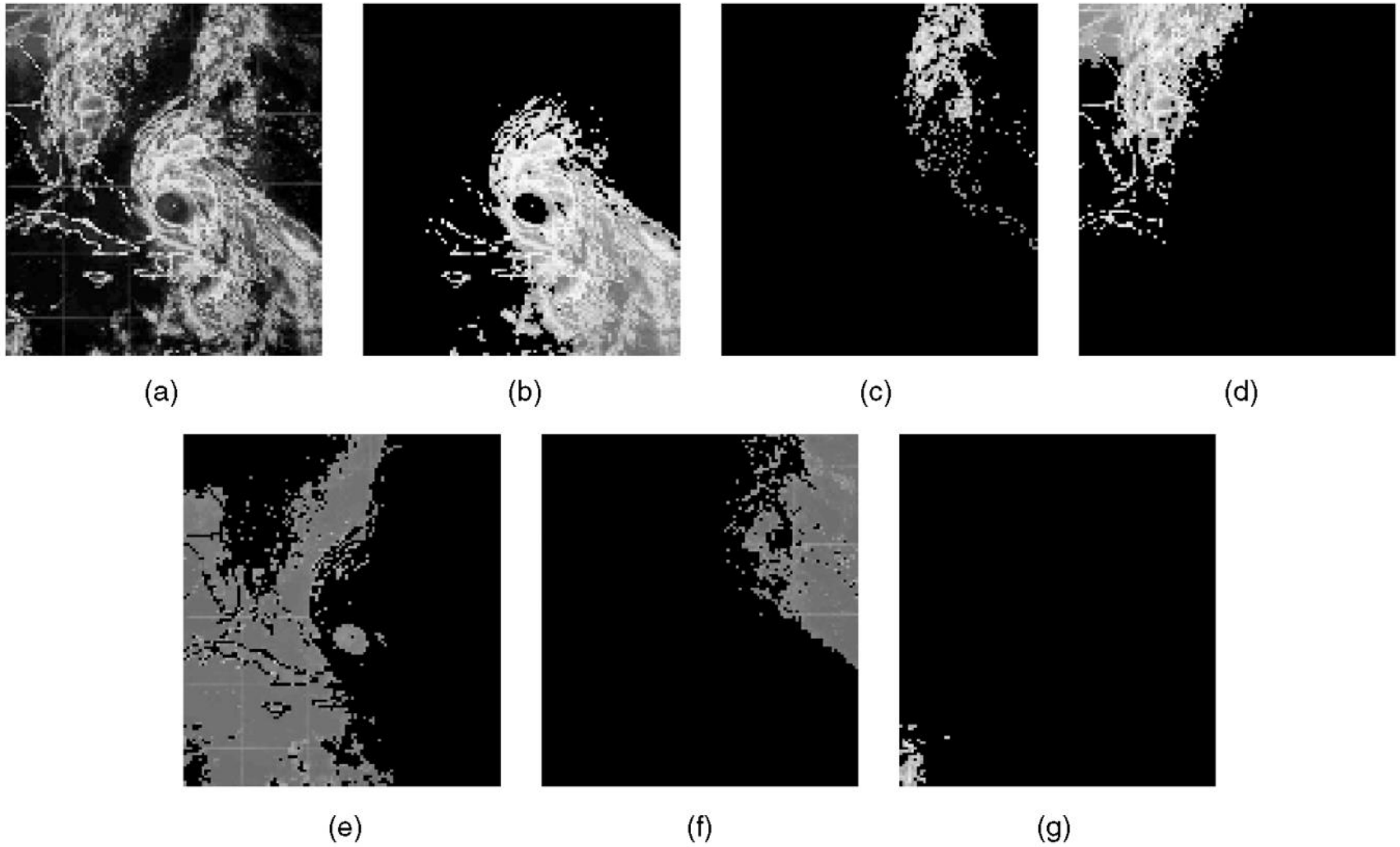
where $X(i)$ is the spatial location of node i, and $F(i)$ is a feature vector based on intensity, color, or texture information.

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Ncut and Image Segmentation



Ncut and Image Segmentation



Application: Unsupervised Word Segmentation

Zheng Yuan, et al. Unsupervised Multi-granular Chinese Word Segmentation and Term Discovery via Graph Partition. *Journal of Biomedical Informatics* (2020), 110:103542.

Dictionary-based methods

- Generally based on the maximum matching principle
- Pros:
  - Fast and accurate with a comprehensive lexicon
- Cons:
  - Not satisfactory when the lexicon coverage is limited (domain text)
- Notable packages:
  - jieba

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Supervised algorithms

- Generally treats word segmentation as a sequential tagging problem
- Common models:
  - Logistic regression, HMM, SVM, conditional random fields, deep learning
- Pros:
  - The most competitive ones in benchmark tests
  - Can use dictionary as features (hybrid)
- Cons:
  - Poor generalizability when applied to a different corpus
  - Requires a large annotated corpus to train
  - Single granularity and biased
- Notable packages:
  - ICTCLAS
  - THULAC
  - PKUSEG

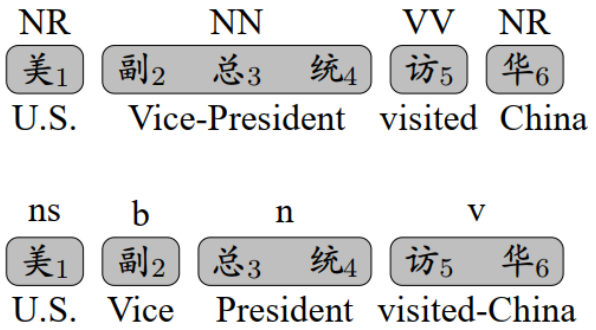


Figure 1: Incompatible word segmentation and POS tagging standards between CTB (upper) and People's Daily (below).

W. Jiang, L. Huang, Q. Liu, Automatic adaptation of annotation standards: Chinese word segmentation and POS tagging - A case study, ACL-IJCNLP 2009 - Jt. Conf. 47th Annu. Meet. Assoc. Comput. Linguist. 4th Int. Jt. Conf. Nat. Lang. Process. AFNLP, Proc. Conf. (2009) 522–530.

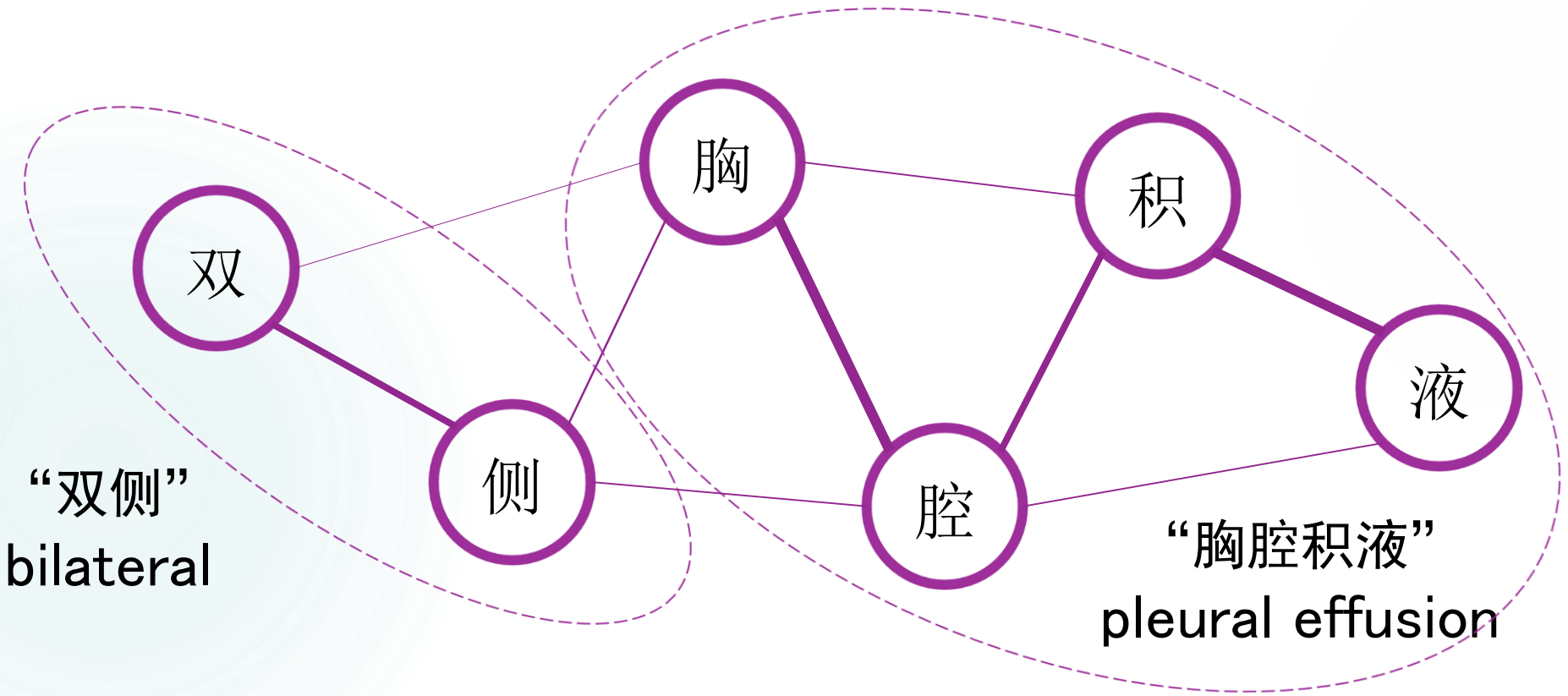
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Unsupervised word segmentation

- Can you think of word segmentation as a clustering analysis?
- Can you think of word segmentation as a graph partition problem?

Think of word segmentation as cutting a graph

Input: 双侧胸腔积液 Bilateral pleural effusion



Designing the edge weights

- $w_{i,i+1} = \alpha \cdot \max\left(P(s_{i+1}|s_i), P(s_i|s_{i+1}), P(s_{i+1}|s_i, s_{i-1}), P(s_i|s_{i+1}, s_{i+2})\right) f(s_i s_{i+1}),$
- $\alpha = \begin{cases} 1, & s_i \notin weaken_set \text{ and } s_{i+1} \notin weaken_set \\ 0.1, & s_i \in weaken_set \text{ or } s_{i+1} \in weaken_set \end{cases}$,
- $weaken_set$ = {未、见、可、或、行、于、及、和、的}
- $f(s_a s_b) = \max(0, \log_2 \#(s_a s_b) - \log_2 5)$,
- $w_{i,j} = 0, |i - j| \neq 1.$

Unsupervised word segmentation

- Word segmentation can be achieved with spectral clustering
  - Clusters = words
  - Ratio cut or normalized cut?
- Only requires corpus statistics – unsupervised
- How to determine the number of clusters (words)?

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Use eigenvalue threshold to control the segmentation

Example segmentation results (with adjustable granularity)
患者 无 咳嗽 、 咳痰 ， 无 咯血 ， 无 发热 等 不适 症状（eig.cut = 0.15）
明确诊断 原发性右肺下叶周围型鳞癌（eig.cut = 0.15） 明确 诊断 原发性 右肺下叶 周围型 鳞癌（eig.cut = 2）
冠状动脉粥样硬化性心脏病（eig.cut = 0.15） 冠状动脉 粥样硬化性 心脏病（eig.cut = 2）
甲状腺功能异常（eig.cut = 0.1） 甲状腺功能 异常（eig.cut = 0.15） 甲状腺 功能 异常（eig.cut = 2）
并给予 吉西他滨 、 顺铂 药物治疗（eig.cut = 0.15） 并 给予 吉西他滨 、 顺铂 药物 治疗（eig.cut = 2）
腹部脐下正中线 可见 手术疤痕（eig.cut = 0.15） 腹部 脐下 正中线 可见 手术 疤痕（eig.cut = 1）

Terms unfound in available dictionaries

肉芽增生，当地医院，雾化吸入，肿大淋巴结，气管支架，支架位置，查房记录，靶动脉栓塞，外压性狭窄，明显减轻，家属拒绝，消融治疗，硅酮支架，光动力治疗，上消化道造影，粘稠分泌物，明确诊断，坏死物质，家属要求，止血治疗，降至正常，饮水呛咳，奥施康定，中央型肺癌，参芪扶正，止痛治疗，相对稳定，中药治疗，人民医院，食管上段，化疗期间，办理出院，射频消融治疗，头孢哌酮舒巴坦，气管管腔，被膜支架，上级医师，微创治疗，复张，冻融治疗，基因检测，食管中段，协助诊治，浅表淋巴结，病理类型，削瘤治疗，自动出院，支架移位，多点注射，胸科医院，肠道菌群，心电监测，突然出现，手术治疗，不典型，心肌酶，心梗三项，低分化鳞癌，剧烈咳嗽，抗痨治疗，对症处理，化疗方案，密切观察，营养支持治疗，评效，食管下段，吉非替尼，恶性病变，肿瘤消耗，肿瘤晚期，丹红，降压治疗，抗感染力度，重症肺炎，耐药菌感染，抗炎治疗，上述治疗，手术疤痕，脱机锻炼，血供丰富，粘膜缺损，诱因出现，摄入不足，病理结果，病理学检查，入量不足，信必，气道湿化，复扫，经皮肺穿刺，体温最高，阿桔片，病变性质，左固有上叶，金薯叶，中分化鳞癌，中国医学，收住入院，凝血活酶时间，胸部增强，稀释痰液，湿化气道，曲霉菌感染，咽后壁，金葡菌感染，一过性，乙肝五项，术前准备，中药汤剂，开胸，下动脉，气管受压，甘油果糖，同步放化疗，气喘症状，真菌治疗，临床路径，硬化性心脏病，压积，气体溢出，安返病房，肉芽样组织，返院，左上肺不张，西黄胶囊，万托林，午后低热，血清骨胶素，固有上叶，悬浮红细胞，超声电导，抽出气体，肺泡灌洗液，返院复查，明显减少，五段血糖，外压，不动杆菌，转复，通便治疗，造血原料，粘膜修复，清理坏死，血气离子，气泡溢出，止血合剂，老年男性，动脉主干，不良性，化疗结束，基底节区，病理分型，肿瘤血供，气管再狭窄，大便次数，动脉夹层，诊疗计划，恶性消耗，四肢肌力，爆发痛，海军总医院，脱水降颅压，扩冠治疗……

Terms unfound in available dictionaries

肉芽增生，当地医院，雾化吸入，肿大淋巴结，气管支架，支架位置，查房记录，靶动脉栓塞，外压性狭窄，明显减轻，家属拒绝，消融治疗，硅酮支架，光动力治疗，上消化道造影，粘稠分泌物，明确诊断，坏死物质，家属要求，止血治疗，降至正常，饮水呛咳，奥施康定，中央型肺癌，参芪扶正，止痛治疗，相对稳定，中药治疗，人民医院，食管上段，化疗期间，办理出院，射频消融治疗，头孢哌酮舒巴坦，气管管腔，被膜支架，上级医师，微创治疗，复张，冻融治疗，基因检测，食管中段，协助诊治，浅表淋巴结，病理类型，削瘤治疗，自动出院，支架移位，多点注射，胸科医院，肠道菌群，心电监测，突然出现，手术治疗，不典型，心肌酶，心梗三项，低分化鳞癌，剧烈咳嗽，抗痨治疗，对症处理，化疗方案，密切观察，营养支持治疗，评效，食管下段，吉非替尼，恶性病变，肿瘤消耗，肿瘤晚期，丹红，降压治疗，抗感染力度，重症肺炎，耐药菌感染，抗炎治疗，上述治疗，手术疤痕，脱机锻炼，血供丰富，粘膜缺损，诱因出现，摄入不足，病理结果，病理学检查，入量不足，信必，气道湿化，复扫，经皮肺穿刺，体温最高，阿桔片，病变性质，[illegible]中分化鳞癌，中国医学，收住入院，凝血活酶时间，胸部增强，稀释[illegible]菌感染，咽后壁，金葡菌感染，一过性，乙肝五项，术前准备，中[illegible]气管受压，甘油果糖，同步放化疗，气喘症状，真菌治疗，临床路径，硬化性心脏病，压积，气体溢出，安返病房，肉芽样组织，返院，左上肺不张，西黄胶囊，万托林，午后低热，血清骨胶素，固有上叶，悬浮红细胞，超声电导，抽出气体，肺泡灌洗液，返院复查，明显减少，五段血糖，外压，不动杆菌，转复，通便治疗，造血原料，粘膜修复，清理坏死，血气离子，气泡溢出，止血合剂，老年男性，动脉主干，不良性，化疗结束，基底节区，病理分型，肿瘤血供，气管再狭窄，大便次数，动脉夹层，诊疗计划，恶性消耗，四肢肌力，爆发痛，海军总医院，脱水降颅压，扩冠治疗……

信必~~可(都保)~~
Symbicort Turbuhaler

Terms unfound in available dictionaries

肉芽增生，当地医院，雾化吸入，肿大淋巴结，气管支架，支架位置，查房记录，靶动脉栓塞，外压性狭窄，明显减轻，家属拒绝，消融治疗，硅酮支架，光动力治疗，上消化道造影，粘稠分泌物，明确诊断，坏死物质，家属要求，止血治疗，降至正常，饮水呛咳，奥施康定，中央型肺癌，参芪扶正，止痛治疗，相对稳定，中药治疗，人民医院，食管上段，化疗期间，办理出院，射频消融治疗，头孢哌酮舒巴坦，气管管腔，被膜支架，上级医师，微创治疗，复张，冻融治疗，基因检测，食管中段，协助诊治，浅表淋巴结，病理类型，削瘤治疗，自动出院，支架移位，多点注射，胸科医院，肠道菌群，心电监测，突然出现，手术治疗，不典型，心肌酶，心梗三项，低分化鳞癌，剧烈咳嗽，抗痨治疗，对症处理，化疗方案，密切观察，营养支持治疗，评效，食管下段，吉非替尼，恶性病变，肿瘤消耗，肿瘤晚期，丹红，降压治疗，抗感染力度，重症肺炎，耐药菌感染，抗炎治疗，上述治疗，手术疤痕，脱机锻炼，血供丰富，粘膜缺损，诱因出现，摄入不足，病理结果，病理学检查，入量不足，信必，气道湿化，复扫，经皮肺穿刺，体温最高，阿桔片，病变性质，左固有上叶，金薯叶，中分化鳞癌，中国医学，收住入院，凝血活酶时间，胸部增强，稀释痰液，湿化气道，曲霉菌感染，咽后壁，金葡菌感染，一过性，乙肝五项，术前准备，中药汤剂，开胸，下动脉，气管受压，甘油果糖，同步放化疗，气喘症状，直营治疗，临床路径，硬化性心脏病，压积，气体溢出，安返病房，[illegible]西黄胶囊，万托林，午后低热，血清骨胶素，固有上叶，[illegible]肺泡灌洗液，返院复查，明显减少，五段血糖，外压，不动杆菌，转复，通便治疗，造血原料，粘膜修复，清理坏死，血气离子，气泡溢出，止血合剂，老年男性，动脉主干，不良性，化疗结束，基底节区，病理分型，肿瘤血供，气管再狭窄，大便次数，动脉夹层，诊疗计划，恶性消耗，四肢肌力，爆发痛，海军总医院，脱水降颅压，扩冠治疗……

金黄色葡萄球菌感染
staphylococcus aureus infection

BIRCH
(balanced iterative reducing and clustering using hierarchies)

Zhang, T., Ramakrishnan, R. and Livny, M., 1996. BIRCH: an efficient data clustering method for very large databases. ACM sigmod record, 25(2), pp.103-114.

BIRCH

- Proposed in 1996 for very large scale database clustering
  - Same year: DBSCAN
- Perform clustering analysis in a linear fashion
  - Does not require calculating dissimilarity between every two samples
  - Does not even require the data set to fit in the main memory
- Received the SIGMOD *Test of Time Award* in 2006

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Clustering Feature and CF Tree

- A clustering feature (CF) is a triple summarizing the information that we maintain about a cluster: $CF = (N, \overrightarrow{LS}, SS)$
  - N : the number of samples in this cluster
  - $\overrightarrow{LS}$: linear sum of samples in this cluster
  - SS: squared sum of samples in this cluster

- CF is used for calculating various measure, such as the cluster center ($\overrightarrow{LS}/N$), radius, average linkage between cluster, etc.

- CF is *additive*
  - The CF of a node is the sum of the CFs of its child nodes.

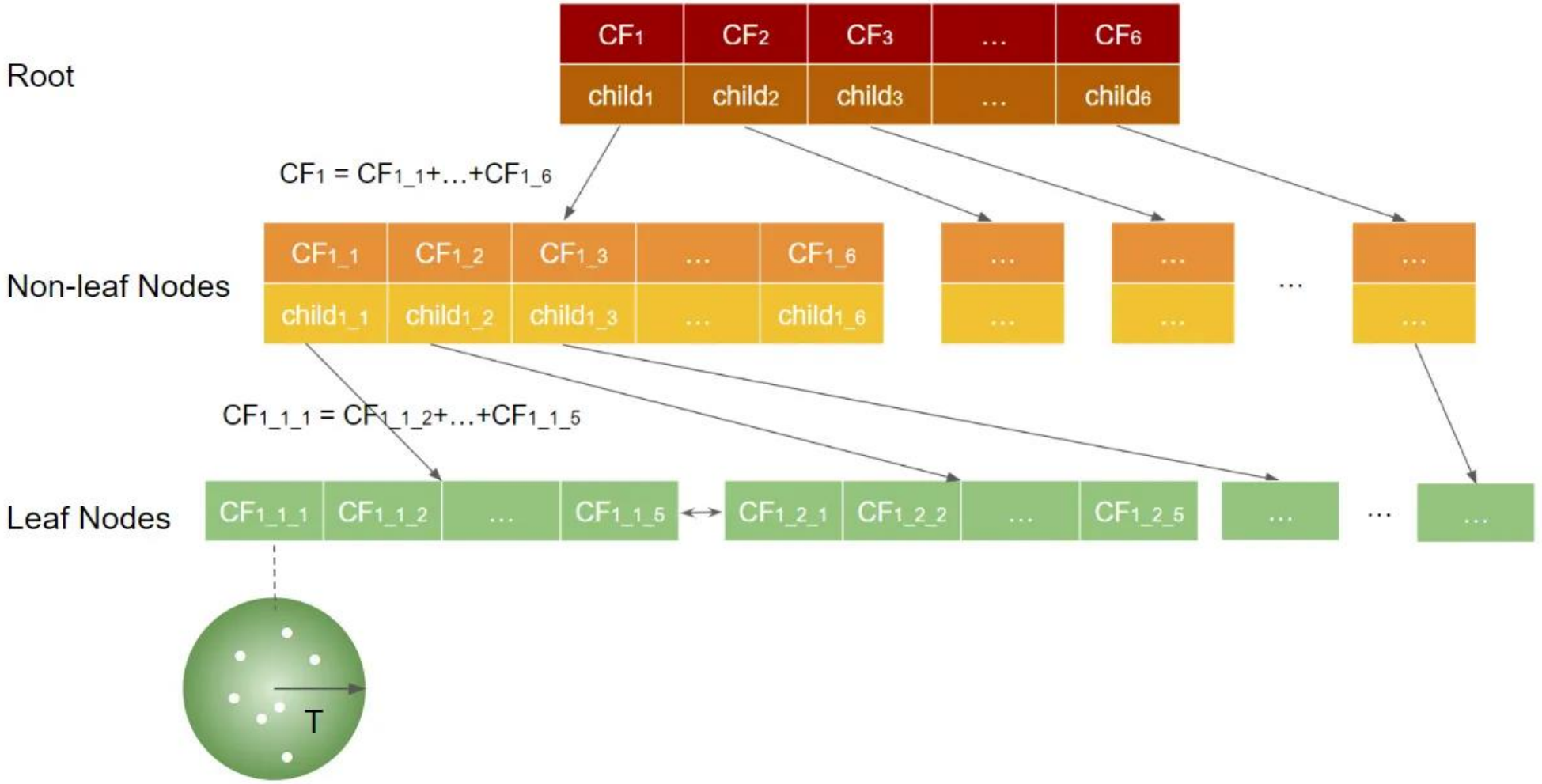
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Clustering Feature and CF Tree

- A CF tree is a height-balanced tree of CFs to easily identify the nearest cluster of a given sample.
- To identify the nearest cluster, iteratively go to the child node with the nearest center.
- The nodes directly containing the clusters (each represented as a CF) are called the leaf nodes.

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Clustering Feature and CF Tree



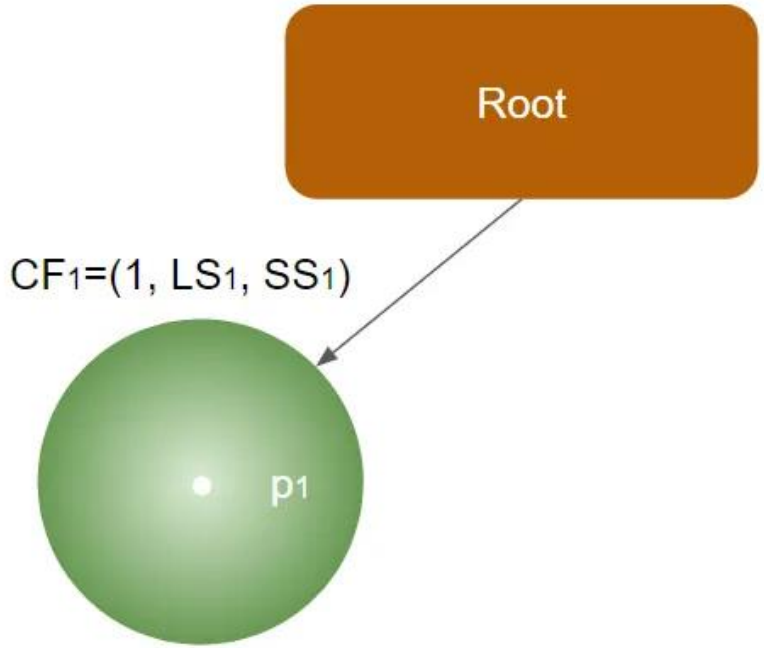
https://medium.com/ai-academy-taiwan/clustering-method-7-fc4c9dd9575b

CF Tree Parameters

- T: max radius allowed in a cluster
- L: max number of children of a leaf node
- B (branching factor): max number of children of a non-leaf node

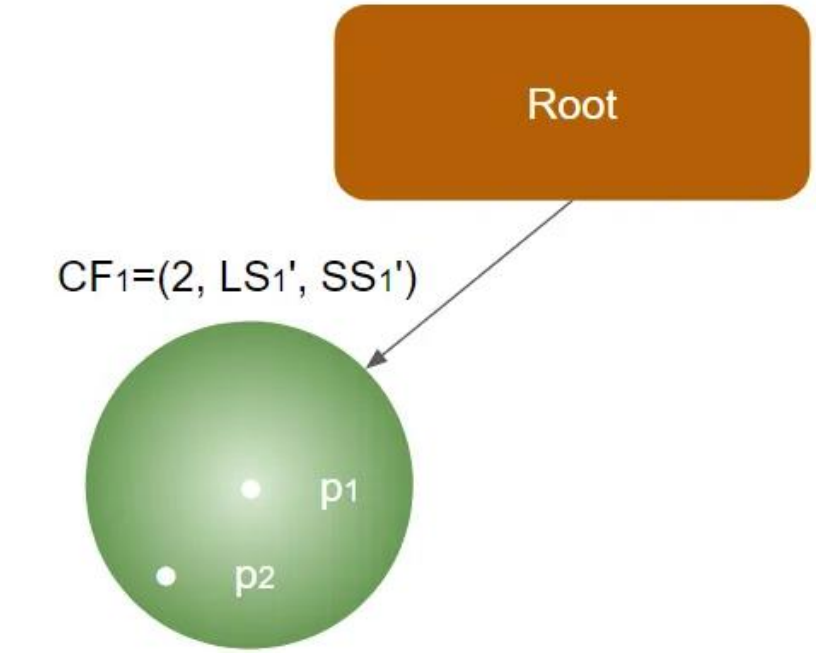
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Constructing the CF tree



Initialize root and insert the first point p1. The root node is now also a leaf node.

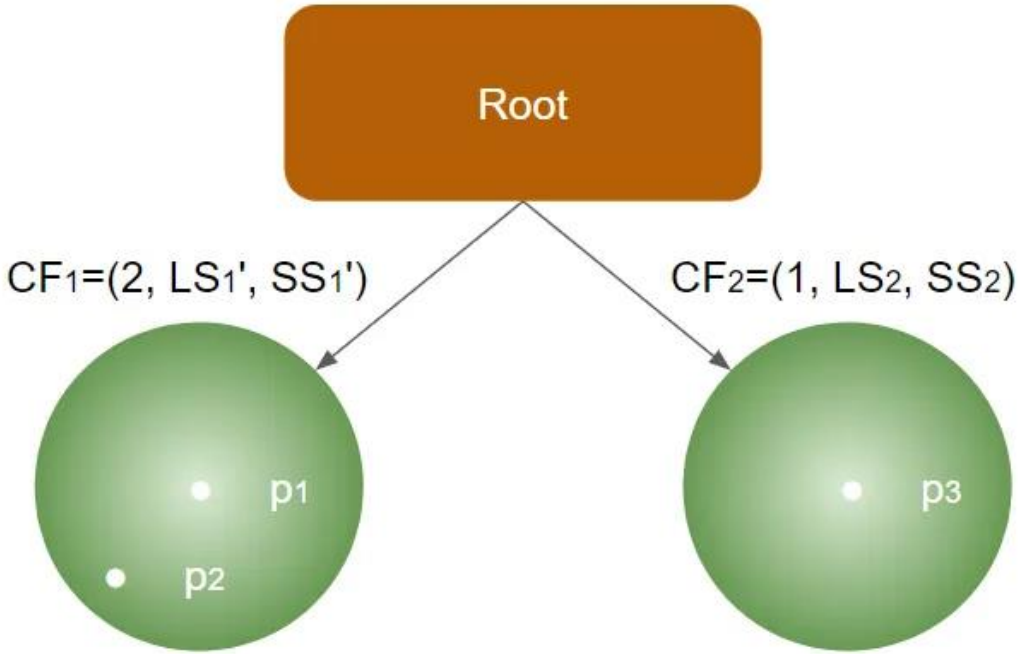
Constructing the CF tree



Insert the second point p2. Its distance from the previous center is less than T and merges to the previous cluster. Note that the CF is updated by the insertion.

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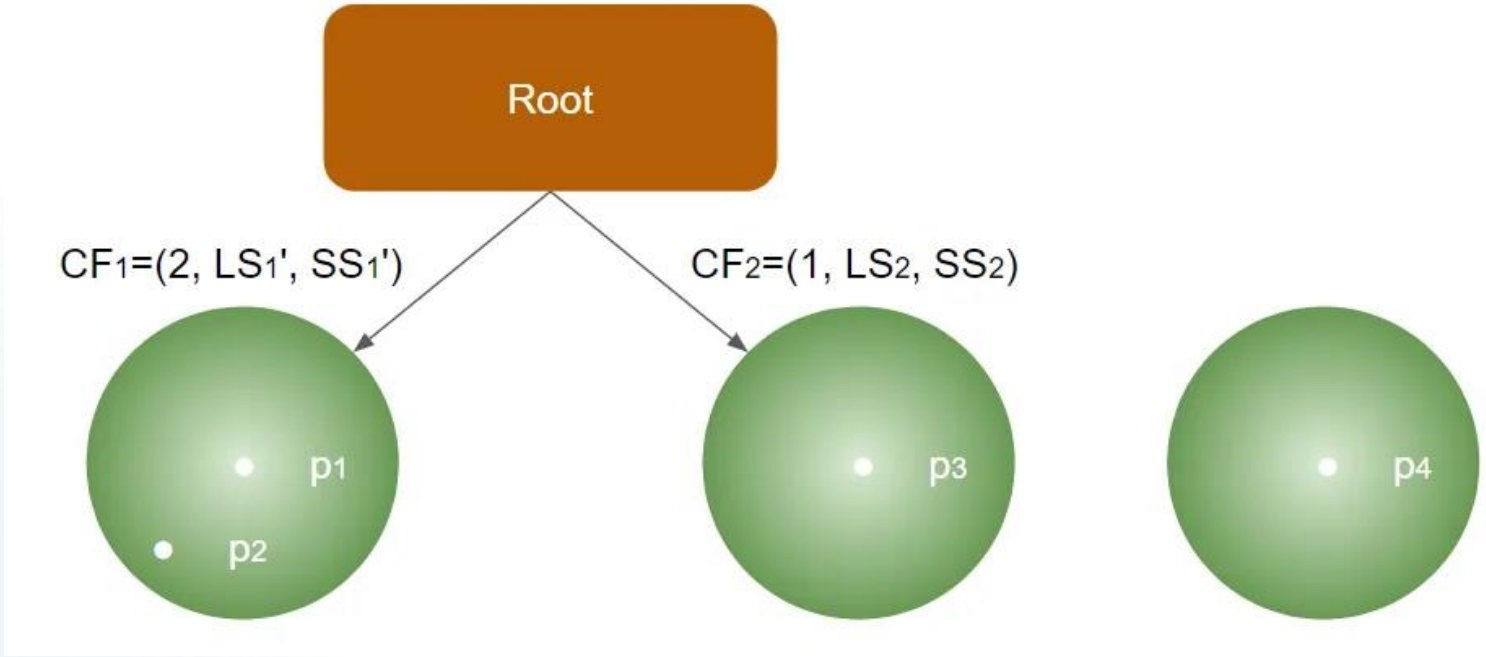
Constructing the CF tree



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Insert the third point p3. If its distance from the previous center is larger than T, a new cluster needs to be created.

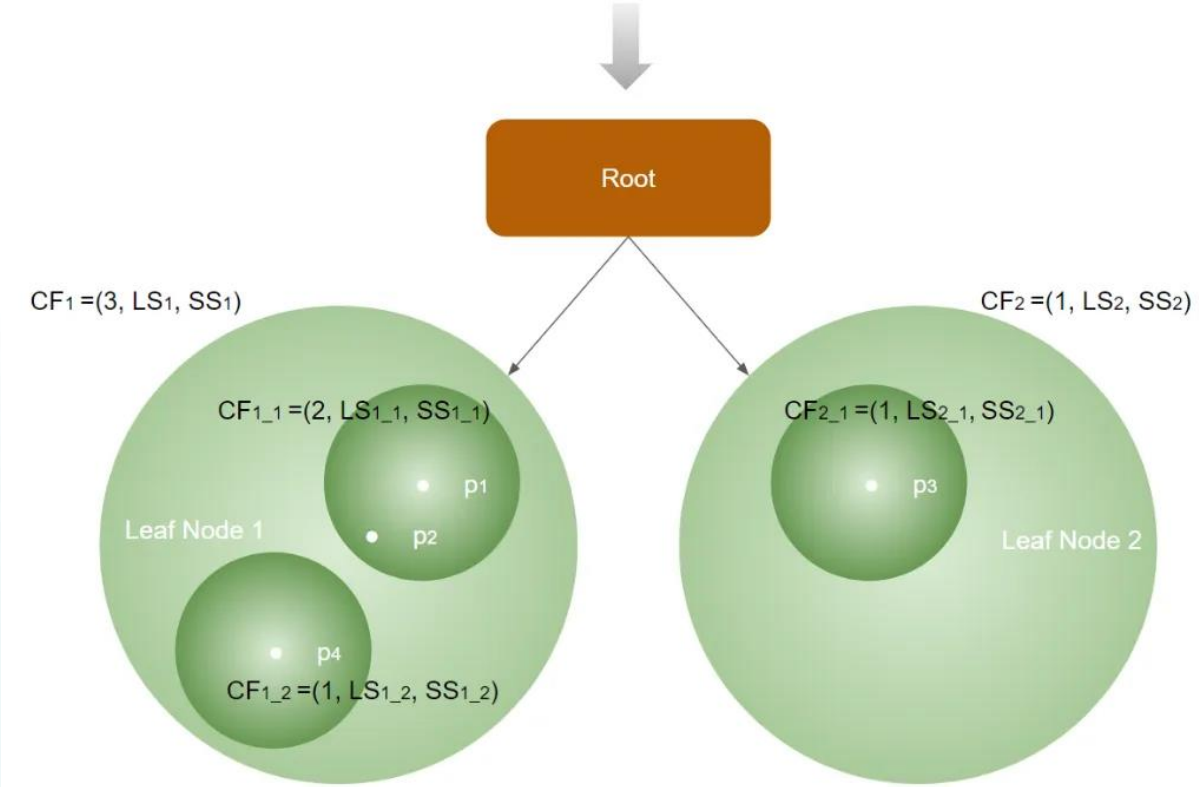
Constructing the CF tree



Insert p4. If it's far from both previous clusters, create a new cluster. Assume L=2, now we need to split their parent (root).

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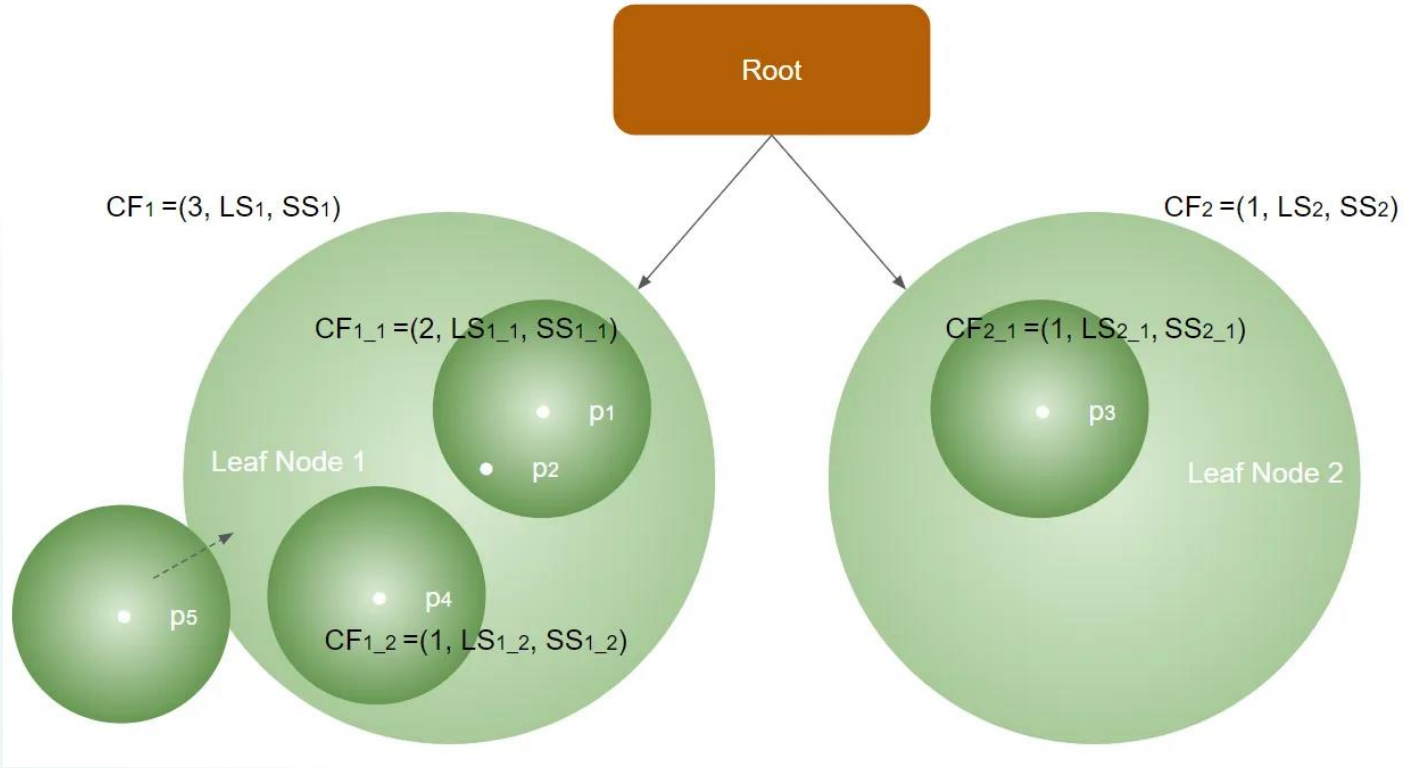
Constructing the CF tree



Node splitting is done by choosing the farthest pair of CFs as seeds, and redistributing the remaining CFs based on the closest criterion. Whenever the root is split, a new root is added on top, and the tree depth grows by 1.

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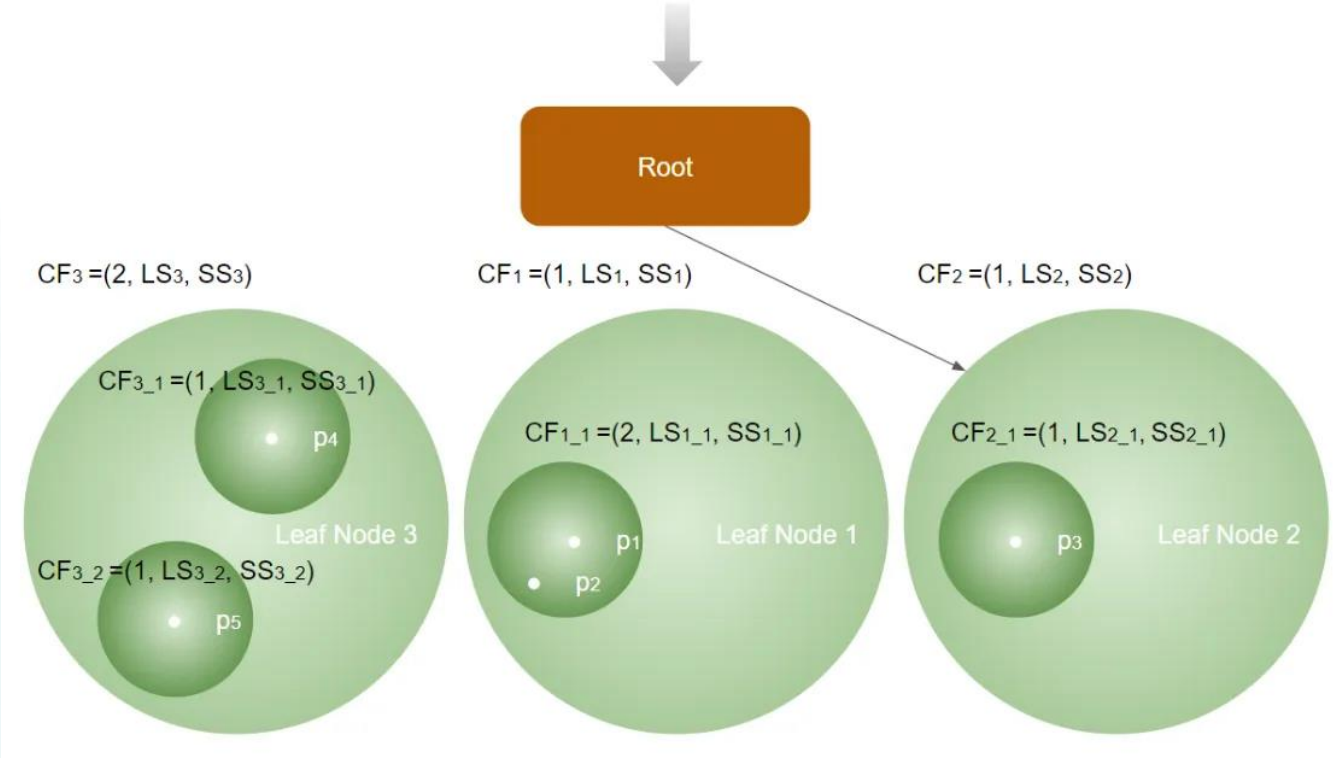
Constructing the CF tree



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Now to add p5, it traverses down the tree, and a new cluster for p5 is to be added at Leaf Node 1. Again, because L=2, Leaf Node 1 needs to be split.

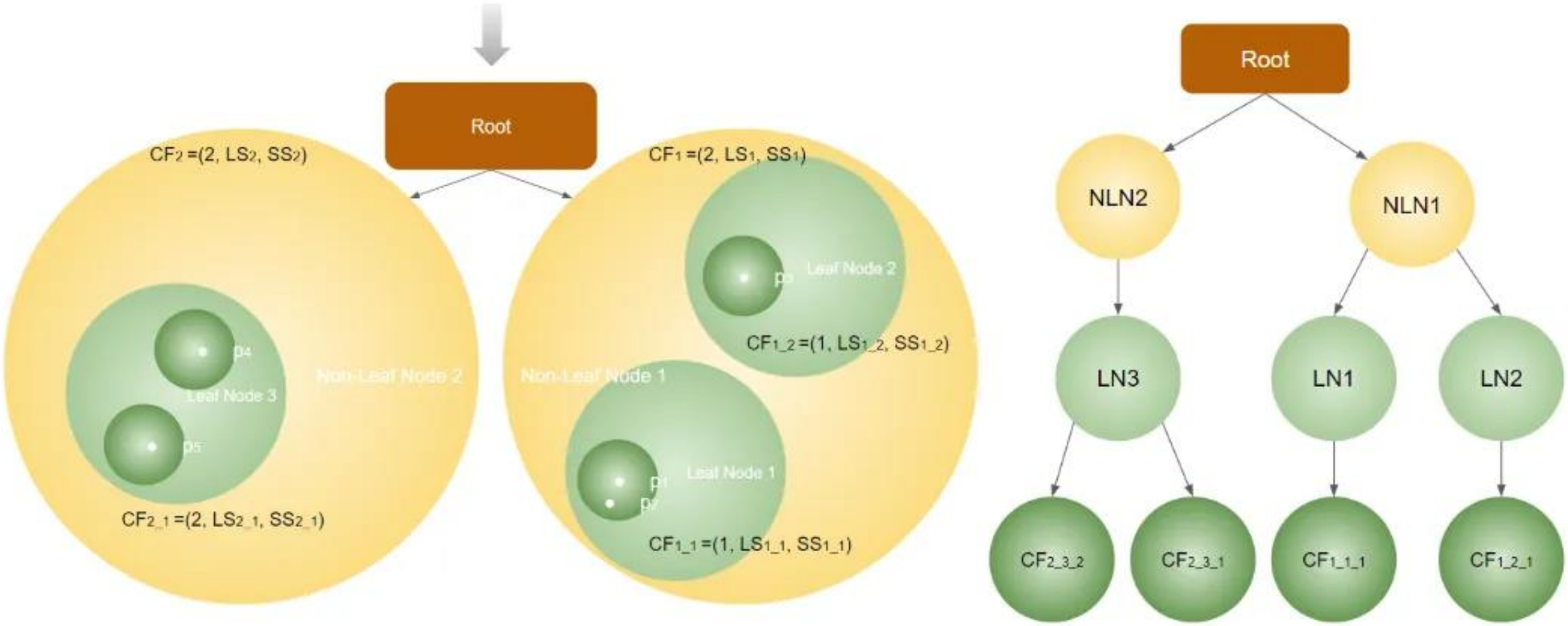
Constructing the CF tree



The old Leaf Node 1 now becomes Leaf Node 3 and new Leaf Node 1. If B=2, this once again requires the root to be split.

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Constructing the CF tree



The previous root was split as two new nodes, and a new root node is added on top. The tree depth once again grows by 1.

Pros and Cons

- Pros:
  - Number of comparison is roughly $O(NB \log_B N)$
  - Very scalable
- Cons:
  - The merging rule based on Euclidean distance can only detect simple patterns such as sphere.
- To compensate, one can use more advanced small data clustering techniques on the leaf nodes after using BIRCH.

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Application: Grouping Synonymous Medical Terms

Ying, H., Zhao, Z., Zhao, Y., Zeng, S. and Yu, S. CoRTEx: Contrastive Learning for Representing Terms via Explanations with Applications on Constructing Biomedical Knowledge Graphs. *Journal of the American Medical Informatics Association* (2024).

Synonym grouping – LLM-assisted BIRCH

- We have 33 million biomedical terms extracted from PubMed. Each term is encoded as a 768-dimensional vector. We want to find all the synonym groups by using a clustering algorithm.

- Calculating all pairwise similarity would require 5.45×10^{14} comparisons. Therefore we use BIRCH in this.

- We modify the insertion criterion of the standard BIRCH:
  - At each branching point, use cosine similarity instead of Euclidean distance.
  - A new point is merged into an existing cluster if a large language model think the new term is synonymous to the cluster members.

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Synonym grouping – LLM-assisted BIRCH


```
"query": "Given one pair containing two terms, you should help me judge whether they refer to the same thing. You should first briefly introduce their meaning in 20 words, then give a judgement use \"yes\" or \"no\" in $\boxed{}$. Finally you should explain why you make this judgement.\nterm1:dental implant therapies\tterm2:dental prosthesis implantation",
"output": "Term 1: Dental implant therapies - Refers to various treatments and procedures involving the placement of dental implants.\nTerm 2: Dental prosthesis implantation - Specifically denotes the process of implanting dental prostheses, such as dentures or bridges.\n\nJudgement: $\\boxed{\\text{Yes}}$\n\nExplanation: Both terms refer to procedures related to dental implants. Term 1 is a broader term that encompasses various therapies involving dental implants, while Term 2 specifically refers to the implantation of dental prostheses. Since dental prosthesis implantation is a type of dental implant therapy, they refer to the same thing."

"query": "Given one pair containing two terms, you should help me judge whether they refer to the same thing. You should first briefly introduce their meaning in 20 words, then give a judgement use \"yes\" or \"no\" in $\boxed{}$. Finally you should explain why you make this judgement.\nterm1:dna methyltransferase spr\tterm2:spr phage methyltransferase",
"output": "Term 1: DNA methyltransferase SPR - Refers to a specific type of DNA methyltransferase enzyme called SPR.\nTerm 2: SPR phage methyltransferase - Specifically denotes a methyltransferase enzyme found in SPR phage.\n\nJudgement: $\\boxed{\\text{No}}$\n\nExplanation: While both terms involve the concept of methyltransferase, Term 1 refers to a specific type of DNA methyltransferase called SPR, whereas Term 2 refers to a methyltransferase enzyme found in SPR phage. These terms do not refer to the same thing as they differ in the specific context and location of the methyltransferase enzyme."
```

术语聚类结果.txt

```
environmental factors|environmental factor
prognostic factor|prognostic factors|prognostics factors
nrf2
increased risk
tumor cellular structure|neoplasm cell|tumor cells|abnormal neoplastic cell|tumorogenic cell|abnormal tumor cell|tumorous cell|neoplastic tumor cell|neoplastic cell|neoplastic abnormal cell|tumour cell|tumor cell type|tumor cell structure|neoplastic cells|tumor cell|neoplastic cellular
microglial cells|microglial cell structure|microglia|microgliacyte|microglial cell
liver cells|hepatocytic cell|hepatoctyte|hepatocytes|hepatocytes-cell|hepatotcyte|hepatocyte|structure of hepatocytes|structure of hepatocyte|hepatic parenchymal cell type|hepatocyt|hepatocytic|hepatic cells|liver parenchymal cell|hepatocyte nos|liver cell|hepatic-cell|hepatic parenchymal cell|cell of the liver|hepatocyte structure|liver-cell
erk1
antiplatelet therapy
anti-inflammatory effect|antiinflammatory effect|antiinflammatory effects
screening tests|screening-test|screening test|screen test|screening-tests|diagnostic screening tests
grey brain matter|gray matter of cortex|gray brain|grey matter of brain|brain gray matter structure|cerebral grey matter|brain matter|brain grey matter|cortex gray matter|brain grey|brain gray matter|gray matter of cerebrum|gray matter of the brain|cerebral cortex gray matter|gray matter of the cerebral cortex|gray matter|brain gray|grey brain|cerebral cortical gray matter|cortical gray tissue|grey matter|structure of gray matter|gray brain matter|encephalic gray matter|cerebral gray|gray matter of brain|gray substance of the brain|cerebral gray matter|gray substance of the cerebrum|gray matter tissue|gray matter of cerebral cortex|matter of gray matter|cortical gray matter|gray matter structure|gray matter of the cerebral hemisphere|gray matter of the cerebrum|grey matter structure|grey matter of the brain
```